

Wide-Bandgap Semiconductor-Based Neuromorphic Computing

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Abstract

Neuromorphic computing has emerged as a promising paradigm to overcome the energy inefficiency and data-transfer bottlenecks of conventional von Neumann architectures by emulating the parallel and adaptive information processing of biological neural systems. To date, most neuromorphic hardware has relied on silicon-compatible or narrow-bandgap materials, which often face intrinsic trade-offs among operating voltage, thermal stability, endurance, and multifunctionality. Wide-bandgap semiconductors (WBGs)-including group III nitrides, gallium oxide, silicon carbide, and diamond-provide an alternative materials platform enabled by their large bandgaps, strong polarization effects, diverse defect states, and compatibility with electronic and optoelectronic device architectures. This review surveys recent progress in WBGs-based neuromorphic computing, with an emphasis on material-enabled device physics rather than isolated demonstrations. Typical device concepts, including memristors, synaptic transistors, and neuronal devices, are systematically discussed together with their underlying resistive switching, charge trapping, polarization modulation, and optoelectronic mechanisms. Strategies for device integration and performance benchmarking are also addressed. Finally, remaining challenges and future research directions toward scalable and energy-efficient neuromorphic systems based on WBGs are outlined.

Keywords: wide-bandgap semiconductors (WBGs), neuromorphic computing, memristor, synaptic transistor.

1. Introduction

The rapid expansion of artificial intelligence has exposed fundamental limitations of conventional von Neumann computing architectures, particularly with respect to energy efficiency and data movement between separated memory and processing units. This data movement overhead primarily stems from the physical separation between memory and processing units, commonly referred to as the memory wall, a critical bottleneck where data frequently needs to be transferred back and forth between these two discrete components and resulting in substantial time latency and excessive energy consumption inherent in the conventional von Neumann framework. Neuromorphic computing, inspired by the brain's massively parallel and event-driven information processing, has therefore emerged as a promising alternative paradigm.^[1] Hardware neuromorphic systems rely on electronic devices capable of emulating synaptic plasticity and neuronal dynamics, which has stimulated extensive research into novel materials and device concepts. To date, most neuromorphic hardware demonstrations have been based on silicon (Si)-compatible technologies or narrow-bandgap semiconductors. However, these material platforms often suffer from trade-offs among endurance, retention, operating voltage, thermal stability, and environmental robustness, limiting their applicability in demanding operating conditions. This has motivated the exploration of alternative material systems that can intrinsically support stable, efficient, and multifunctional neuromorphic behavior.

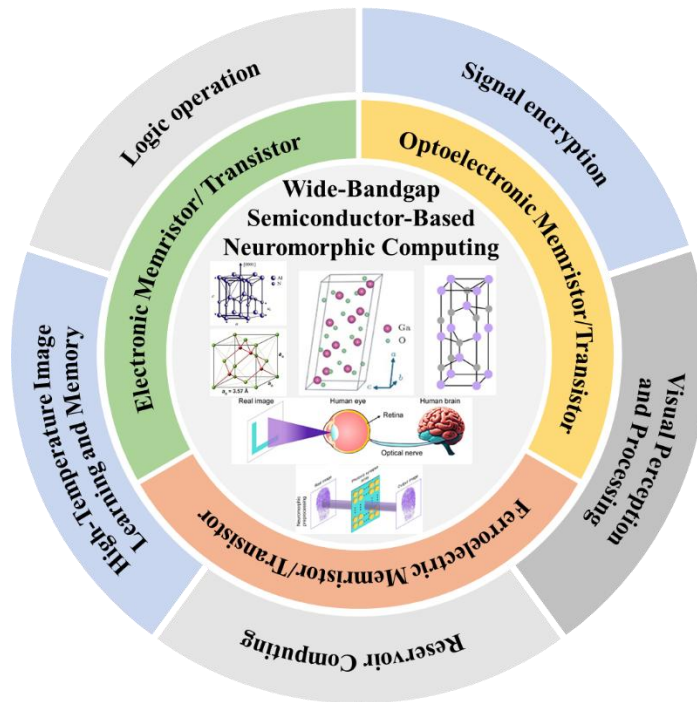


Figure 1. Overview of Wide-Bandgap Semiconductor-Based Neuromorphic Computing. Reproduced with permission^[2]. Copyright 2025, Journal of Materials Science. Reproduced with permission^[3]. Copyright 2019, Materials Today Physics. Reproduced with permission^[4]. Copyright 2025, journal of Materials Research and Technology. Reproduced with permission^[5]. Copyright 2025, Advanced Materials. Reproduced with permission^[6]. Copyright 2024, Carbon-Based Nanofillers and Their Rubber Nanocomposites.

Wide-bandgap semiconductors (WBGs), typically defined by bandgaps larger than $\sim 2.5\text{-}3.0$ eV, are well

established in power electronics and optoelectronics owing to their high breakdown fields, large optical phonon energies, and excellent chemical and thermal stability. These intrinsic attributes also suggest compelling advantages for neuromorphic computing, particularly for devices operating at high-speed, high-power density, or under harsh environments. Importantly, WBGs exhibit rich defect landscapes, strong polarization effects, and versatile interface properties, which can be deliberately exploited to realize analog resistive switching (RS), tunable conductance modulation, and diverse forms of synaptic plasticity.

As the fundamental building blocks of neuromorphic hardware, the performance of synaptic and neuronal devices is governed by the physicochemical properties of their constituent materials. Conventional neuromorphic devices are predominantly based on transition-metal oxides, organic materials, or two-dimensional materials (2DMs).^[1,7] However, these systems are often vulnerable to performance degradation under extreme conditions such as high temperature or intense radiation, limiting their use in aerospace, nuclear, and other specialized applications. In contrast, WBGs offer a distinctive combination of ultra-low leakage current, high electron mobility (10^2 - 10^3 $\text{cm}^2\cdot\text{V}^{-1}\cdot\text{s}^{-1}$), excellent thermal conductivity (up to ~ 400 $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$), and superior thermal and radiation stability. In addition, their favorable compatibility with established CMOS processes enables scalable fabrication and system-level integration. Currently, WBGs explored for neuromorphic computing mainly include group III nitrides (Gallium nitride (GaN), aluminum nitride (AlN), and boron nitride (BN)), gallium oxide (Ga_2O_3), silicon carbide (SiC), and diamond. These materials exhibit unique advantages in synaptic weight (SW) modulation, low-power operation, and extreme-environment adaptability, collectively establishing WBGs as a promising materials platform for the diversified development of neuromorphic computing.

In this review, we focus on neuromorphic computing enabled by several aforementioned representative wide-bandgap semiconductors (WBGs), with particular emphasis on the correlations among materials properties, device architectures, physical mechanisms, and system-level functionalities (Figure 1). Section 2 introduces key WBGs systems and their properties relevant to neuromorphic functions. Section 3 reviews representative neuromorphic device concepts based on WBGs. Section 4 discusses the physical mechanisms underlying synaptic and neuronal behaviors. Section 5 addresses integration strategies, benchmarking metrics, and emerging application scenarios. Finally, section 6 outlines the remaining challenges and future perspectives for WBGs-based neuromorphic computing.

2. Wide-Bandgap Semiconductors for Neuromorphic Devices

2.1 III-V Nitrides: GaN, BN, and AlN

Group III nitride WBGs are characterized by strong spontaneous and piezoelectric polarization, high electron mobility, and mature epitaxial growth technologies, making them a cornerstone of wide-bandgap semiconductor research. GaN, AlN, and BN are three representative III-V nitrides that have been extensively investigated for high-power electronics and deep-ultraviolet (DUV) optoelectronic applications. Beyond these conventional roles, their unique electronic structures, polarization effects, and interface properties provide a fertile platform for polarization-controlled synaptic transistors, memristive interfaces, and optoelectronic neuromorphic devices.

As a core material of third-generation semiconductor technologies, GaN exhibits a wide bandgap of ~ 3.4 eV, high thermal conductivity, and excellent resistance to high temperature, radiation, and chemical corrosion, collectively supporting stable device operation under extreme conditions and establishing GaN as a promising candidate for memristive devices.^[8] Moreover, its hexagonal wurtzite crystal structure endows GaN with superior optoelectronic properties, enabling direct coupling between electrical and optical stimuli for visual neuromorphic

systems.^[9] As illustrated in Figure 2(a), wurtzite GaN is a non-centrosymmetric single-crystal material with a well-defined polar axis (*c*-axis), along which strong spontaneous polarization and anisotropic piezoelectric coupling are present. This spontaneous polarization arises from the relative displacement between the charge centers of Ga³⁺ cations and N³⁻ anions, caused by the non-centrosymmetric stacking of Ga and N atomic planes along the polar axis. The [0001] crystallographic direction corresponds to Ga-polar surfaces, whereas the [000 $\bar{1}$] direction corresponds to N-polar surfaces, leading to polarity-dependent electronic and interfacial behaviors.^[10]

AlN exists in both wurtzite and zinc blende crystal structures, with the thermodynamically stable phase under ambient conditions being the wurtzite structure, characterized by lattice constants of $a = 3.112 \text{ \AA}$ and $c = 4.982 \text{ \AA}$, as shown in Figure 2(b).^[11] Similar to GaN, wurtzite AlN exhibits strong spontaneous polarization along the *c*-axis due to the relative displacement of Al and N sublattices, resulting in two opposite polarization states associated with Al-polar and N-polar orientations. However, direct switching between these polarization states requires electric fields exceeding the dielectric breakdown strength, rendering conventional ferroelectric domain switching inaccessible. In strained AlGaN alloys, lattice distortion along the *c*-axis induces additional atomic displacement, giving rise to pronounced piezoelectric polarization effects that further modulate internal electric fields.^[12] Notably, scandium (Sc) doping in AlN introduces substantial lattice deformation, manifested as a reduced *c/a* ratio and increased metal-nitrogen bond parameter u along the *c*-axis.^[13] With increasing Sc concentration, the crystal structure evolves from the stable wurtzite phase toward a metastable layered hexagonal structure characteristic of ScN. Although this layered hexagonal phase is intrinsically nonpolar, it represents an intermediate structural state between two oppositely polarized wurtzite configurations. Under this condition, external electric fields can induce polarization switching in AlScN ferroelectric thin films, enabling RS behavior.

BN represents a distinct class of III-V nitrides as a two-dimensional wide-bandgap material. In contrast to graphene, BN exhibits excellent insulating properties, ultra-low dielectric loss, and exceptionally high thermal conductivity exceeding $750 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$.^[14,15] These attributes have enabled high-performance ultraviolet photodetectors with responsivities exceeding 10^5 A/W , ultra-low- k dielectrics for sub-5 nm interconnect technologies, and resistive memory devices with endurance up to 10^8 cycles.^[16–18] Among various BN polymorphs, hexagonal boron nitride (h-BN) is the most widely studied, featuring a layered structure analogous to graphite with atomically smooth interfaces and excellent chemical stability.^[15] As shown in Figure 2(c), its two-dimensional (2D) crystal structure and wide bandgap facilitate charge trapping and tunneling modulation, enabling memristive behavior and lossless integration of ultra-thin, high-performance neuromorphic devices.

Overall, group III-V nitride WBGSs integrate multiple advantageous properties, including wide bandgaps ranging from ~ 3.4 to 6.2 eV , high thermal conductivity, high electron mobility, and exceptional chemical, thermal, and radiation stability, as listed in Table 1. Compared with conventional oxide materials, nitride semiconductors offer distinct advantages such as reduced phase randomness, strong and tunable polarization effects, and superior compatibility with established semiconductor processes. These characteristics help suppress thermally induced fluctuations and random filament formation during RS, thereby improving device uniformity and computational accuracy. Furthermore, the intrinsic coupling between spontaneous polarization and piezoelectric effects provides new degrees of freedom for regulating ion migration pathways, opening promising avenues for low-power, nonvolatile neuromorphic devices. Combined with their relatively low defect density, tunable trap states, and compatibility with Si substrates, III-V nitrides offer a robust and scalable materials platform for high-performance neuromorphic computing.

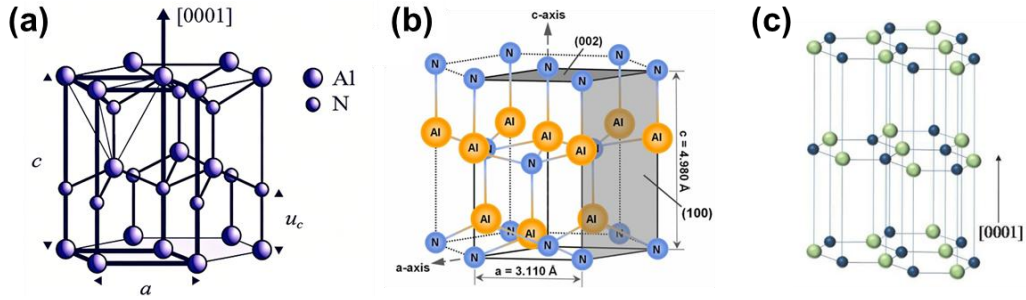


Figure 2. Crystal structures of (a) GaN, Reproduced with permission ^[2]. Copyright 2025, Journal of Materials Science. (b) AlN, Reproduced with permission ^[19]. Copyright 2024, Journal of Alloys and Compounds. and (c) BN. Reproduced with permission ^[20]. Copyright 2017, Published by Advanced Electronic Materials.

2.2 Silicon Carbide (SiC)

SiC is a representative WBGS that has attracted sustained interest due to its close structural and technological compatibility with silicon-based electronics. As a binary compound composed of silicon and carbon in a 1:1 atomic ratio, SiC exhibits more than 200 polymorphic and heteromorphic crystal configurations. Among these, 3C-, 4H-, and 6H-SiC are the most technologically relevant polytypes (Figure 3). ^[21] In particular, 4H-SiC has emerged as a preferred material for neuromorphic device fabrication owing to its high electron mobility ($\sim 1000 \text{ cm}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}$), wide bandgap (3.2-3.4 eV), and excellent thermal stability, with a melting point approaching 2700 °C. SiC also exhibits a thermal conductivity approximately 3.3 times higher than that of conventional Si, enabling efficient heat dissipation and stable operation under high power density. Furthermore, its favorable tunability for both n-type and p-type doping provides flexibility in device design. The recent availability of commercial 8-inch 4H-SiC wafers represents a major milestone, laying a solid foundation for large-scale integration and manufacturing of SiC-based neuromorphic circuits. SiC devices can operate reliably at temperatures up to ~ 550 °C and exhibit superior resistance to ionizing radiation, making them highly suitable for neuromorphic computing in harsh environments. From a device-physics perspective, SiC offers additional advantages for neuromorphic applications. A high-quality SiO₂ layer can be readily grown on the SiC surface, providing effective oxidation protection and enabling well-controlled interfaces. Both intrinsic and extrinsic defects in SiC, including vacancies and interface traps, can act as charge trapping centers, facilitating nonvolatile and analog RS behaviors. These characteristics enable the realization of synaptic plasticity and stable weight modulation, positioning SiC as a robust materials platform for high-temperature and radiation-hardened neuromorphic systems.

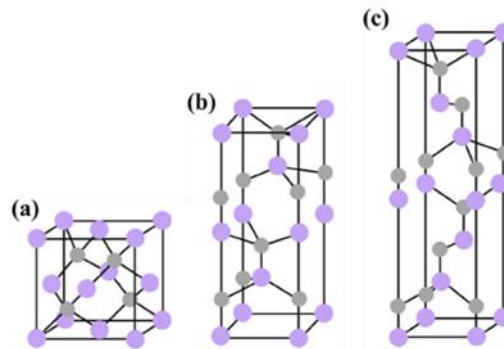


Figure 3. Crystal structures of (a) 3C-, (b) 4H-, and (c) 6H-SiC polytypes. Reproduced with permission ^[4]. Copyright 2025, journal of Materials Research and Technology.

2.3 Gallium Oxide (Ga_2O_3)

Ga_2O_3 is a typical ultra-wide-bandgap semiconductor (UWBGs), featuring a bandgap energy ranging from ~ 4.6 to 4.9 eV and existing in six distinct crystalline phases (α , β , γ , δ , ϵ , and κ) [22], as shown in Figure 4. Among them, β - Ga_2O_3 has been most extensively studied due to its superior thermodynamic stability and relatively mature fabrication technologies. Ga_2O_3 is characterized by an exceptionally high breakdown electric field ($\sim 8 \text{ MV}\cdot\text{cm}^{-1}$), excellent chemical stability, and optical transparency, making it attractive for high-power and optoelectronic applications (Table 1). Thin films of Ga_2O_3 can be deposited using a variety of techniques, including magnetron sputtering and atomic layer deposition, and exhibit good compatibility with flexible substrates. As an emerging UWBGs, Ga_2O_3 has demonstrated broad application potential in power electronics, including resistive random-access memory (RRAM), Schottky barrier diodes, field-effect transistors, solar-blind photodetectors, radiation detectors, and DUV optoelectronic devices. [23] Ga_2O_3 exhibits annealing-temperature-dependent phase transitions and structural evolution, which provide additional degrees of freedom for tailoring its electronic and RS behaviors. In neuromorphic applications, Ga_2O_3 -based memristors have demonstrated fast switching speeds, long data retention, and excellent endurance, enabling low-energy, multi-level, and scalable synaptic devices compatible with complementary metal-oxide-semiconductor (CMOS) processes and three-dimensional (3D) architectures. [24] Moreover, Ga_2O_3 exhibits strong ultraviolet photoresponse arising from oxygen-vacancy-related defect states, enabling photo-electric synergistic modulation of SW. These characteristics make Ga_2O_3 particularly attractive for optoelectronic neuromorphic systems and visual bionic applications. [3] While its ultra-high breakdown field and defect-rich physics provide clear advantages for memristive neuromorphic devices, challenges related to thermal conductivity and heat dissipation remain and must be addressed for high-density integration.

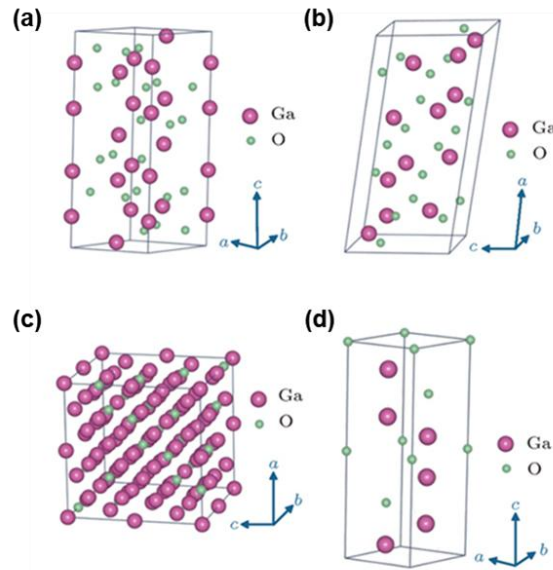


Figure 4. Crystal structures of (a) α -, (b) β -, (c) γ - and (d) ϵ - Ga_2O_3 . Reproduced with permission [3]. Copyright 2019, Materials Today Physics.

2.4 Diamond

Diamond represents a unique class of UWBGs, combining an exceptionally wide bandgap with ultra-low electronic noise, which enables highly sensitive detection of optical signals. [25] This characteristic is particularly

advantageous for neuromorphic photonic systems, where enhanced wavelength sensitivity and image recognition accuracy are required. ^[26] Additionally, diamond exhibits extreme hardness, outstanding chemical stability, and strong radiation resistance, ensuring long device lifetimes and reliable operation under harsh environmental conditions (Figure 5). ^[27] Its ultra-high thermal conductivity further supports stable device performance at elevated temperatures, making diamond an attractive candidate for neuromorphic applications in extreme environments. ^[28] Despite these advantages, diamond-based neuromorphic devices face notable limitations. First, their intrinsic photoresponse is primarily confined to the DUV region, which does not match the spectral sensitivity of biological vision systems. Second, the volatile nature of diamond photocurrents poses challenges for optical information storage and nonvolatile neuromorphic functions. Recent studies have shown that these limitations can be mitigated through interface engineering, particularly by exploiting hybrid diamond/graphene heterostructures. At the diamond/graphene interface, intermediate carbon bonding configurations (sp^3 - sp^2) can reversibly transition between insulating and conductive states under external stimuli such as electric fields or optical excitation. This dynamic interfacial modulation gives rise to tunable resistance states and photoinduced conductivity, enabling the emulation of synaptic plasticity and neuronal behaviors. ^[29] Overall, diamond and related ultra-wide-bandgap materials offer compelling opportunities for neuromorphic computing in extreme environments and photonic systems. Continued progress in interface engineering, defect modulation, and heterogeneous integration is expected to further unlock their potential for robust and multifunctional neuromorphic devices.

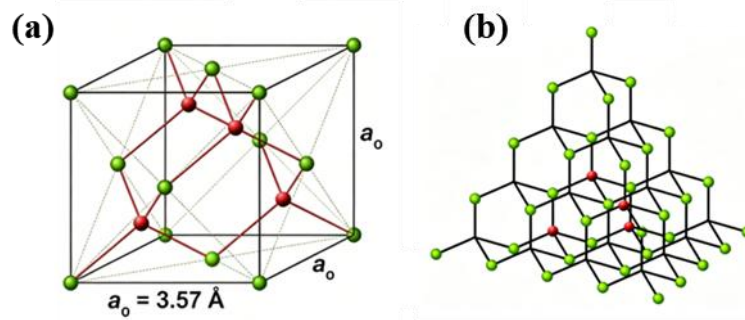


Figure 5. Structure of diamond (a) unit cell structure. (b) 3D structure. Reproduced with permission ^[6]. Copyright 2024, Carbon-Based Nanofillers and Their Rubber Nanocomposites.

Table 1. A comparative discussion of band gap, thermal conductivity, breakdown field, relative permittivity, and mobility. ^[22,23,30–34]

Properties	Unit	GaN	AlN	h-BN	4H-SiC	6H-SiC	3C-SiC	β -Ga ₂ O ₃	Diamond
Band gap	eV	3.4	6.2	5.8	3.26	3.00	2.20	4.2-4.9	5.5
Thermal conductivity	W/cm·k	1.1	3.2	3.9	3	2.4	1.2	0.23[010] 0.13[100]	10
Breakdown field	MV/cm	3.3	2	7	4.9	4.9	4.9	8	10
Relative permittivity	-	9	8.5	6.85 (in-plane) 5.06 (out-of plane)	9.66	9.66	9.72	10	5.5
Mobility	cm ² /Vs	1200	135	35	800	400	900	300	2000

Notes: - represent not reported

3. Device Architectures for WBG Neuromorphic Computing

3.1 Performance Parameters

3.1.1 Memristor

As the fundamental building blocks of neuromorphic computing and next-generation nonvolatile memory systems, the practical performance of memristors is governed by a set of interrelated parameters that are intrinsically determined by their underlying RS mechanisms. Most memristors adopt a metal-insulator-metal (MIM) sandwich structure, within which RS behavior arises from electrically induced physical or chemical changes in the active layer or at the electrode interfaces. ^[35]

Based on their dominant switching physics, memristors can be classified into three principal categories, ^[24] electrochemical metallization (ECM), valence change mechanism (VCM), and thermochemical mechanism (TCM), as schematically illustrated in Figure 6. In memristive operation, the SET process corresponds to the transition from a high-resistance state (HRS) to a low-resistance state (LRS), whereas the RESET process restores the device from the LRS back to the HRS. In conductive-filament-based ECM and VCM devices, an external electric field drives the migration of metal cations or oxygen vacancies, leading to the formation of conductive filaments during the SET process, while reverse bias or Joule heating induces filament rupture during RESET. In contrast, redox-based or interface-type memristors rely on oxidation-reduction reactions at the electrode/active-layer interface, where modulation of the Schottky barrier height governs the resistance state. These distinct switching mechanisms directly dictate the achievable performance metrics of memristive devices.

Reliable RS with clearly distinguishable HRS and LRS, typically characterized by an on/off ratio exceeding 10^2 for practical applications, forms the foundation for effective information storage and signal encoding. ^[36,37] For neuromorphic computing, gradual and controllable conductance modulation is essential to emulate SW updates. Such analog switching behavior is closely linked to the controllability of ion migration, defect redistribution, or charge trapping dynamics inherent to the RS mechanism. Energy efficiency is another critical parameter, particularly for large-scale crossbar arrays, where the target energy consumption per synaptic event ideally approaches the femtojoule (fJ) level comparable to biological synapses. This energy cost is fundamentally determined by the activation energy required for conductive filament formation/rupture or interfacial redox reactions. Fast switching speed, spanning from sub-nanosecond to microsecond regimes, is necessary for high-throughput information processing and depends on the migration kinetics of ions or charge carriers in the active layer. Long-term reliability, reflected by high endurance (exceeding 10^9 switching cycles) and robust data retention (over 10 years at room temperature or stable operation at elevated temperatures), is particularly critical for WBGs-based devices intended for harsh environments, and is governed by the structural stability of conductive paths and the reversibility of the switching processes.

Additionally, device-to-device and cycle-to-cycle uniformity is a prerequisite for high-precision neuromorphic computation, as variability directly translates into computational noise in large-scale arrays. Uniformity is strongly influenced by the homogeneity of the active material, interface quality, and the degree of control over the RS mechanism. Consequently, optimizing memristor performance requires coordinated material engineering strategies-such as defect modulation, doping, heterostructure design, and interface engineering-to precisely regulate the dominant switching mechanisms. Continued advances along these directions are essential for translating memristor-based neuromorphic devices from laboratory demonstrations toward scalable, reliable, and commercially viable technologies.

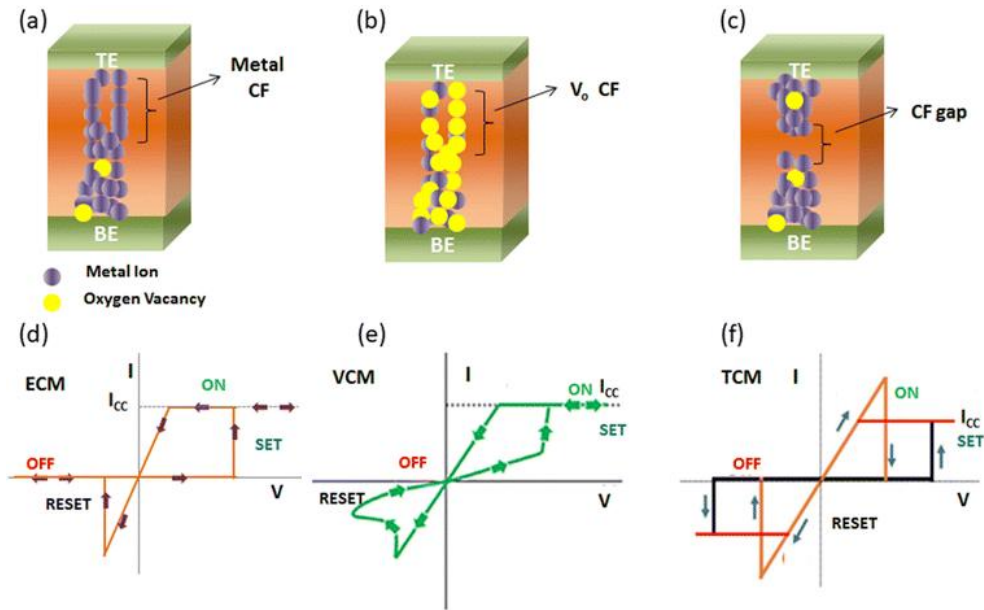


Figure 6. Schematic illustrations of RS mechanisms. (a) ECM, (b) VCM, and (c) TCM. Representative hysteretic current-voltage (I-V) characteristics for (d) ECM, (e) VCM, and (f) TCM devices. Reproduced with permission [24]. Copyright 2025, Nanoscale.

3.1.2 Synaptic devices

The nervous system of organisms consists of a vast number of neurons interconnected by synapses, where presynaptic and postsynaptic neurons communicate through synaptic structures to facilitate the conduction of nerve impulses between neurons. [38] Synaptic connections are primarily categorized into two types: chemical synapses and electrical synapses. [39] Chemical synapses dominate in mammals, with signal transmission mediated by neurotransmitters, while electrical synapses rely on direct electrical signal conduction and are more prevalent in fish and amphibians. As schematically illustrated in Figure 7, a chemical synapse comprises three core structural components: the presynaptic terminal (axon terminal of the presynaptic neuron), the synaptic cleft, and the postsynaptic terminal (dendrite or cell body of the postsynaptic neuron). [40] Upon the arrival of an action potential at the presynaptic terminal, membrane depolarization induces calcium ion influx through voltage-gated channels, triggering the release of neurotransmitters into the synaptic cleft. [41] These neurotransmitters bind to receptors on the postsynaptic membrane, modulating ion channel activity and generating postsynaptic currents or potentials. [42] Depending on the neurotransmitter type and receptor response, synaptic transmission can be excitatory or inhibitory, producing excitatory postsynaptic currents/potentials (EPSC/EPSP) or inhibitory postsynaptic currents/potentials (IPSC/IPSP). [43,44] These two complementary modes jointly regulate neural signal propagation and information processing. Artificial synaptic transistors aim to replicate these fundamental behaviors by mapping SW to channel conductance and using gate stimuli to emulate presynaptic spikes.

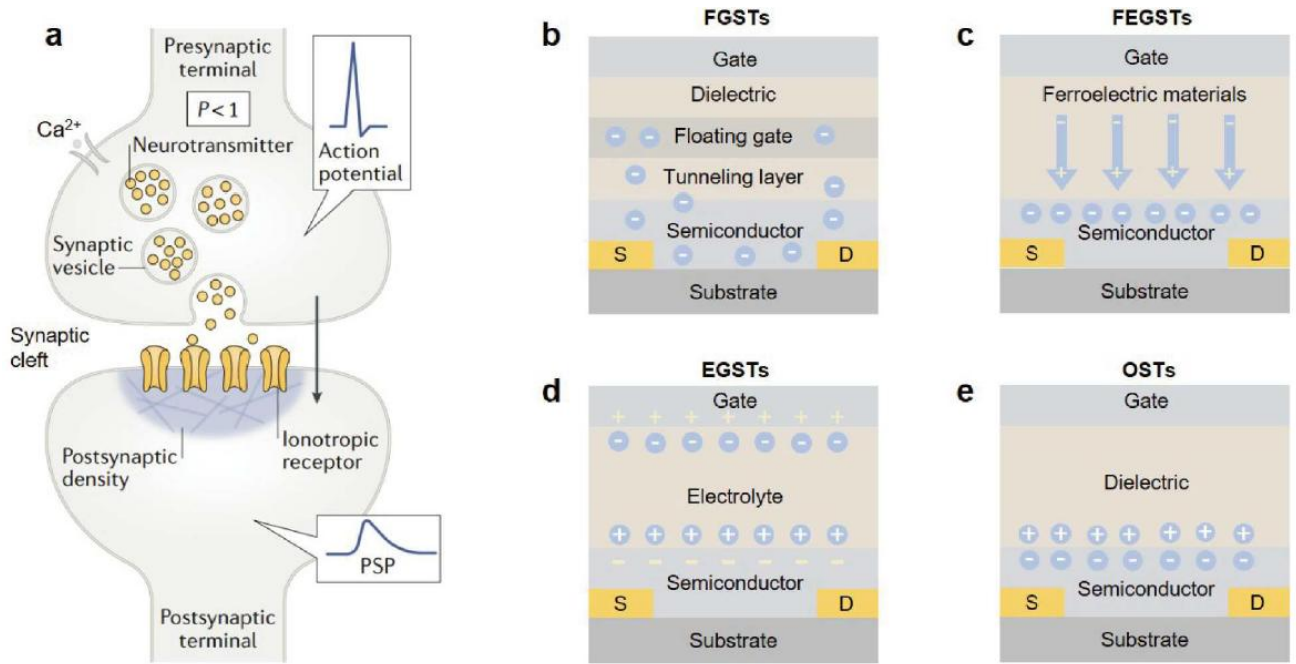


Figure 7. Working mechanism of synaptic transistors. (a) The structure of a chemical synapse. Reproduced with permission ^[45]. Copyright 2019, Nature Reviews Neuroscience. The typical structure of (b) floating gate, (c) ferroelectric, (d) Electrolyte, (e) optoelectronic synaptic transistor. Reproduced with permission ^[46]. Copyright 2025, Science China Materials.

SW quantifies the strength of neuronal connections, while synaptic plasticity describes the adaptive modulation of SW in response to stimuli. ^[47] Artificial synaptic transistors can emulate essential plasticity characteristics of biological synapses, including short-term plasticity (STP) and long-term plasticity. ^[48,49] STP is commonly manifested as paired-pulse facilitation (PPF), or paired-pulse depression (PPD), where the postsynaptic response to a second stimulus is enhanced or suppressed relative to the first. ^[50,51] The PPF/PPD ratio is typically quantified as $100 \times (A_2/A_1)$, where A_1 and A_2 denote the response amplitudes of successive stimuli. Long-term plasticity typically manifests as either long-term potentiation (LTP) or long-term depression (LTD), which are induced and regulated by synaptic plasticity rules such as spike-timing-dependent plasticity (STDP) or spike-rate-dependent plasticity (SRDP). ^[52,53] Transitions from STP to LTP can be achieved by adjusting stimulation parameters such as pulse amplitude, duration, or frequency. In addition, synaptic transistors can reproduce dynamic learning-forgetting-relearning processes, a hallmark of adaptive intelligence.

Energy efficiency is a critical performance metric for synaptic transistors, particularly for large-scale neuromorphic systems. For three-terminal devices, synaptic energy consumption is commonly estimated as $E = V \times I \times t$, where V is the source-drain voltage, I is the postsynaptic current, and t is the stimulus duration. ^[54] Recent advances have reduced the energy consumption of synaptic transistors to the femtojoule (1-10 fJ) level per synaptic event-comparable to, or even below, that of biological synapses. Such energy-efficient operation, combined with precise weight tunability and high endurance, positions WBG-based synaptic transistors as promising building blocks for scalable neuromorphic computing, adaptive memory systems, and low-power intelligent electronics.

3.2 Memristors and Two-terminal Synaptic Devices

Memristors are recognized as the fourth fundamental circuit element, complementing resistors, capacitors, and

inductors, as first theoretically proposed by Chua in 1971. [37] The defining characteristic of a memristor is that its resistance state can be continuously modulated by the history-dependent flow of electrical charge through the active medium, resulting in nonlinear and nonvolatile RS behavior. Owing to these features, memristors have been widely adopted as artificial synapses for emulating biological learning and memory functions. [55] To date, most reported memristive devices rely on electrical stimuli to regulate conductance states. While electrically driven memristors have undergone decades of development and achieved substantial progress, optoelectronic memristors remain at a comparatively early stage. Depending on the mode of conductance modulation, optoelectronic memristors can be broadly classified into photo-electric cooperative devices, in which optical and electrical stimuli jointly regulate resistance states, and fully optical memristors, where conductance modulation is driven primarily by light. These emerging device concepts are particularly attractive for neuromorphic vision and in-sensor computing. WBGs-based memristors typically exploit defect migration, interfacial redox reactions, and either filamentary or interface-type RS. [56–58] Compared with conventional oxide memristors based on narrow-bandgap materials, WBGs memristors generally exhibit superior thermal stability, enhanced endurance, and more robust operation under high electric fields and extreme environmental conditions, making them well suited for harsh-environment neuromorphic applications.

3.2.1 Electronic devices

a) GaN

To date, most GaN-based electrical memristors have been realized using GaN microwires (MWs) or nanowires (NWs), where low-dimensional geometries amplify defect, surface, and polarization effects. In 2016, Pan and co-workers developed a piezoelectric memristor with multilevel switching behavior based on bamboo-like GaN MWs. [59] By coupling the piezoelectric and semiconducting properties of GaN, a piezotronic modulation mechanism was introduced: mechanical deformation induces strain-driven piezoelectric polarization at defect boundaries, which effectively modulates the internal potential landscape and reduces the SET voltage. Building on this concept, a flexible piezoelectronic sensory memory device (SMD) based on bamboo-like GaN MWs was later demonstrated. [60] In this device, external strain programs the device into a HRS, while electrical stimuli erase it back to a LRS. The strain-induced piezopotential triggers the dissolution and redistribution of nitrogen-vacancy-based conductive channels, enabling spatially resolved mapping of tactile stimuli and memory storage. Such piezoelectronic memory behavior provides a direct pathway for converting mechanical sensory information into electrical synaptic signals, highlighting the potential of GaN for interactive and multimodal neuromorphic systems. Beyond one-dimensional GaN structures, heterostructure engineering has further expanded GaN memristor functionality. Ji *et al.* [61] reported a hybrid-dimensional memristor consisting of wrinkled MoS₂-wrapped GaN NWs with a spray-coated Ag NW network as the top electrode. The bipolar RS behavior is governed by the local electric fields generated by the wrinkled morphology, which drive oxygen ion migration along the NWs surface to achieve memristive functionality. Separately, Huang *et al.* [62] fabricated bipolar GaN NW memristors via wet etching using H₃PO₄. Surface chemistry modulation alters adsorbed molecules and dangling bonds, enabling electric-field-driven accumulation or depletion of oxygen vacancies and surface electrons. These studies collectively demonstrate that GaN memristors can exploit polarization effects, surface defect dynamics, and heterointerfaces to emulate complex synaptic functions. The performance comparisons in nitride-based memristors are listed in Table 2.

Table 2. Performance comparisons in nitride-based memristors.

Memristors	Power consumption (pJ)	Memory window (s)	Forming voltage (V)	Retention memory window (s)	Endurance (Cycles)	ON/OFF ratio	Ref.
GaN MW	-	-	-	-	> 282	-	[59]
GaN MW	-	-	-	$> 3 \times 10^2$	> 100	$\sim 10^4$	[60]
Ag/GaN MW/Ag	-	-	-	$> 10^5$	> 150	10^3	[63]
MoS ₂ @GaN NW	-	3.4×10^3	-	3.4×10^3	798	$\sim 10^3$	[61]
H ₃ PO ₄ -treated GaN NW	-	3×10^3	0	$> 3 \times 10^3$	> 100	10^3	[62]
TiN/AlN/TiN	-	-	5, 2.5	3×10^5	10^5	> 10	[64]
TiN/AlN/Pt	-	-	-	-	-	10^2	[65]
TiN/AlN/Pt	-	-	7.8	$> 10^4$	100	-	[66]
Ag/AlN/TiN Ni/Ag/AlN/TiN	1.15	-	-	10^4	-	-	[67]
Al/AlN/TiN	-	-	4	10^4	> 120	$> 10^4$	[68]
TiN/AlN/Pt	-	-	1.51	10^6	-	> 10	[69]
Ag/ AlN	-	-	-	10^4	100	> 10	[70]
Au/Al/Al ₂ O ₃ /AlN/ITO Au/Al/ AlN/ITO	-	-	5	10^4	20-100	10^2 - 10^3	[71]
Pt/Al/TaO _x /AlN/Al ₂ O ₃ /Pt	-	-	4	$> 10^4$	100-500	> 100	[72]
Pt/HfO _x /AlN/TiN	-	-	-	$> 10^4$	500	-	[73]
Pt/Al/TiO _x /HfO ₂ /AlN/Pt	-	-	4.16	$> 10^4$	100	-	[74]
Ag/AlN/AlScN/AlN/Pt	0.2	-	1.15	-	-	10^4	[75]
Metal/ScAlN/n-GaN	-	-	-	10^5	1000	10^5	[76]
Cu/Sc _{0.2} Al _{0.8} N/Pt/Ti	-	-	-	$> 10^4$	-	10^4	[77]
Au/Ti/h-BN/Au	2	1.7	-	-	> 600	> 50	[78]
Au/Ti/h-BN/Gr(E)	-	-	0	-	> 200	$\sim 10^7$	[79]
Ni/h-BN/Ni	-	-	-	-	> 600	$\sim 10^7$	[80]
Ag/PMMA/h-BN/Cu	-	-	-	10^5	> 800	$\sim 10^5$	[81]

b) AlN

A series of MIM-type AlN-based memristors have been demonstrated via the adoption of diverse electrode materials, including Pt, Cu, Al, Ag, Pd, and TiN. In 2013, Chen *et al.* [82] reported that complementary RS behavior was achieved in both TiN/AlN/Cu/AlN/TiN ECM cells and Pt/AlN/TiN/AlN/Pt ionic RS systems. Copper atoms readily undergo redox reactions in numerous semiconductors and insulators, these redox processes induce copper filament deposition, which preferentially occurs along grain boundaries or other crystalline defects of the AlN thin

film. The rupture and reconnection of copper filaments trigger the HRS via thermal release and LRS via cold release effects of the device, respectively. For the Pt/AlN/TiN/AlN/Pt device, both Pt and TiN electrodes are electrochemically inert and do not inject metal cations into the AlN film. Thus, a high-voltage electroforming process was employed to activate the ionic switching effect of the system, enabling memristive functionality. In 2016, Choi *et al.* [65] fabricated an Al/AlN/TiN-based memristor featuring ultra-fast switching time and relatively low switching current. They revealed that oxidation occurs at the top interface of the AlN layer, forming an AlO_xN_y phase and Al-rich conductive channels. The switching behavior of the device is dominated by the migration of nitrogen vacancies. More recently, Ling *et al.* [69] also developed an Al/AlN/TiN memristor with an ultra-thin (~ 3.3 nm) AlN RS layer based on the same mechanism, achieving a retention time of 10^6 s. Furthermore, memristors based on the formation/rupture mechanism of Ag-based conductive filaments have been extensively explored. Kwon *et al.* [67] developed Ag/AlN/TiN memristors, investigating their long-term and short-term memory characteristics as well as I-V curves. Additionally, they studied the switching and conduction mechanisms of Ni/Ag/AlN/TiN devices using conductive filament schematics and energy band diagrams, opening new avenues for AlN-based memristors in neuromorphic systems (Figure 8a). Mohanty *et al.* [70] prepared AlN/Ag-based conductive-bridge random-access memories (CBRAMs), where the insertion of a Ta layer effectively controlled the diffusion of Ag atoms in the switching layer. Beyond single-component AlN systems, heterojunction engineering with oxides has been employed to tune memristor performance via the synergistic regulation of oxygen and nitrogen vacancies. Guo *et al.* [71] introduced a thin Al_2O_3 layer on the surface of the AlN film. The device performance is dominated by the nitrogen vacancy conductive filament mechanism, and the Al_2O_3 layer stabilizes the conductive path by providing additional oxygen vacancies, thereby enhancing device performance in memory and computing applications. Recently, Choi *et al.* [72] embedded a 1.2 nm-thick AlN layer between the TaO_x switching layer and the Al_2O_3 tunnel barrier layer, which acts as a built-in current limiter. This design facilitates controlled filament formation and achieves intrinsic adaptability without the need for external circuits. The resulting Pt/Al/ TaO_x /AlN/ Al_2O_3 /Pt RRAM device enables polarity-dependent dual-mode switching within a single cell. Similarly, the Pt/Al/ TiO_x /HfO₂/AlN/Pt device fabricated by Park *et al.* [74] exhibits abrupt filamentary RS behavior, whose characteristics are strongly dependent on the variation of defect states in the TiO_x layer. A natural oxidation reaction between the TiO_x oxygen reservoir and the Pt/Al top electrode forms an AlO_x layer serving as an overshoot suppression layer (OSL). Moreover, an ultra-thin AlN layer at the interface between the HfO₂ layer and the Pt/Ti bottom electrode acts as an oxygen barrier layer (OBL), effectively inhibiting oxygen ion migration.

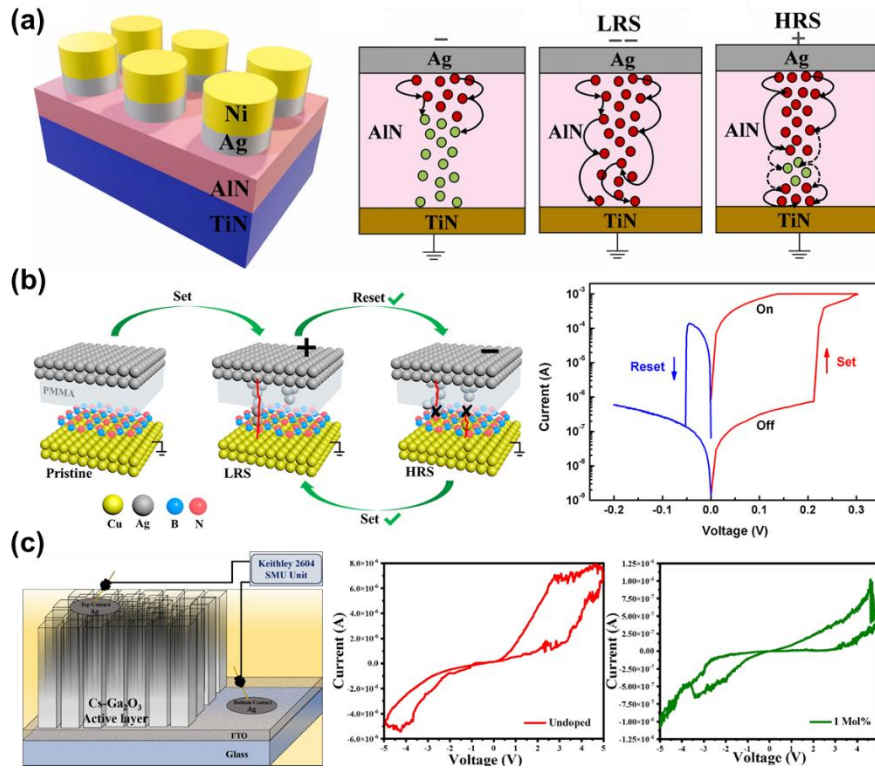


Figure 8. Memristors and their working mechanism of (a) Ag/AlN/TiN, Reproduced with permission [67]. Copyright 2022, Journal of Alloys and Compounds. (b) Ag/PMMA/h-BN/Cu, Reproduced with permission [81]. Copyright 2021, Materials & Design. and (c) cesium-doped Ga₂O₃ (Cs-Ga₂O₃). Reproduced with permission [83]. Copyright 2022, Memories Materials Devices Circuits Systems.

c) BN

Within the BN, amorphous BN (*a*-BN) and h-BN have been extensively employed for memristor. Similar to other nitride-based memristors, the MIM architecture is predominantly adopted for BN-based devices, where the formation/rupture of CFs remains the dominant physical mechanism underlying non-volatile RS. Currently, MIM-structured BN memristors typically consist of an active metal top electrode (e.g., Ag, Cu), a BN functional layer, and an inert metal bottom electrode (e.g., Pt, Au), featuring high spatial utilization and integration density. *a*-BN exhibits a disordered atomic arrangement without long-range order. Its ultra-wide bandgap (>6 eV) and low dielectric constant ($\kappa \approx 1.8$) render it an ideal candidate for conventional memristor architectures. [84] Khot *et al.* [16] demonstrated a CMOS-compatible Ag/*a*-BN/Pt memristor tailored for neuromorphic applications. By modulating the current compliance (I_{cc}) of the *a*-BN functional layer, multi-level RS was achieved, enabling multi-bit storage capability with 10^4 endurance cycles and a non-volatile retention time exceeding 10^6 s while maintaining distinct resistance state discrimination. In contrast, h-BN comprises vertically stacked 2D planes interconnected via van der Waals (vdW) forces. Its atomic-scale thickness and exceptional insulating properties make it an ideal platform for investigating interfacial charge dynamics and quantum mechanical tunneling effects in nanoscale memristive systems. Intrinsic lattice defects (e.g., vacancy clusters, grain boundaries) in CVD-synthesized h-BN serve as preferential pathways for metal ion migration from electrodes, facilitating controllable CF formation via ECM. Völkel *et al.* [80] developed a bilayer Ni/h-BN/Ni crossbar memory, achieving an on/off ratio exceeding 10^7 at operating voltages below 3 V. Teja Nibhanupudi *et al.* [78] designed an Au/Ti/h-BN/Au memristor with a multi-layer 2D structure, leveraging the intrinsic defect tolerance of CVD-grown h-BN to realize ultra-fast switching speeds below 5 ns. To optimize device performance, heterostructure engineering and interface modulation have been

widely explored. Ge *et al.* ^[81] fabricated an Ag/PMMA/h-BN/Cu structured device by inserting a PMMA interfacial polymer layer between the monolayer h-BN and the top electrode (Figure 8b). This design effectively restricts CF dimensions and suppresses excessive CF formation on the bottom Cu electrode, endowing the device with low operating voltage, high on/off ratio, and long retention time. When combined with carbon-based materials, carbon ions form carbon conductive filaments, thereby achieving memristive properties. Xie *et al.* ^[79] developed vertical h-BN memristors with graphene edge contacts. Then, several graphene/hBN heterostructured memristors, i.e., G/h-BN/G/SiO₂/Si ^[85], Ti/MLG/h-BN/MLG/Au ^[86], Au/Ti/G/h-BN/G/Au ^[87] were proposed. Suzuki *et al.* ^[88] successfully grew h-BN layers on both single carbon nanotubes and carbon nanotube bundles via dry spinning technology, and fabricated h-BN/CNT heterostructure memristor devices. The device's resistive behaviour originates from the disordered amorphous carbon structure between h-BN layers, wherein carbon atoms penetrate the hexagonal BN layers to form conductive pathways, inducing a transition from a HRS to a LRS.

d) SiC

Within the SiC family, amorphous SiC (*a*-SiC), β -SiC, and 4H-SiC have been exploited for memristor fabrication. SiC-based memristors typically rely on electrical pulse-induced metal ion migration to trigger the formation and rupture of metallic CFs within the SiC active layer. This process efficiently modulates device conductivity, endowing the devices with memristive characteristics. In 2011, Lee *et al.* ^[89] first reported an *a*-SiC-based RRAM. Employing sputtered SiC as the insulating layer and Cu as the active electrode, the device achieved 10⁵ endurance cycles and a two-order-of-magnitude on/off ratio by incorporating a germanium antimony telluride (GeSbTe) barrier layer. Zhong *et al.* ^[90] demonstrated a Cu/*a*-SiC/Au memristor with an ultra-high on/off ratio of 10⁹. Chen *et al.* ^[91] introduced a titanium antimony telluride (TiSbTe) layer into *a*-SiC_{0.11}:H memristors, which enhanced the switching uniformity of the devices. This improvement was attributed to the formation of Si dangling bond nanochannels within the TiSbTe layer. For single-crystalline SiC devices, Liu *et al.* ^[92] simultaneously achieved threshold switching (TS) and bipolar RS behaviors in a single Ag/4H-SiC/Pt device by tuning the I_{cc} . Qin *et al.* ^[93] proposed a full electron beam evaporation approach to fabricate Cu/SiC/Pt structures on 2-inch intrinsic Si substrates, realizing the transition from volatile to non-volatile switching. The device exhibited low symmetric switching voltages of ± 0.5 V, >200 cycling endurance, >10³ s retention time, an on/off ratio of $\sim 10^3$, and at least six stable resistance states. Kapur *et al.* ^[94] developed a memristor based on back-end-of-line (BEOL)-compatible SiC, demonstrating excellent bipolar switching behavior with an on/off ratio of 10⁴. A 3×3 artificial synaptic array constructed from these devices successfully emulated synaptic functions corresponding to the “learning-forgetting-relearning” process. Recently, the same group fabricated SiC/Si bilayer resistive memory via plasma-enhanced chemical vapor deposition (PECVD) ^[95]. Precise modulation of resistance states and multi-level switching were achieved by varying pulse conditions, with the device exhibiting over 10⁹ endurance cycles at an on/off ratio of $\sim 10^2$. Santra *et al.* ^[96] reported a bipolar RS device with low power consumption and high performance based on a ZnO@ β -SiC composite, which eliminated the need for a pre-forming process. Embedded Zn₂SiO₄ nanocrystals within the amorphous matrix enabled device stabilization up to 10⁴ cycling endurance and 10⁴ s retention time, governed by the CF formation and dissolution mechanisms driven by oxygen vacancies and metal cations. The performance comparisons in SiC-based memristors are listed in Table 3.

Table 3. Performance comparisons in SiC-based memristors.

Memristors	Power consumption (pJ)	Memory window (s)	Forming voltage (V)	Retention memory window (s)	Endurance (Cycles)	ON/OFF ratio	Ref.
TiN/GST/Cu/SiC/Pt	10 ⁵ ~10 ⁶	-	-	> 10 ⁴	> 10 ⁴	> 10 ³	[89]
Cu/ α -SiC/Au	-	-	-	-	-	10 ⁹	[90]

Memristors	Power consumption (pJ)	Memory window (s)	Forming voltage (V)	Retention memory window (s)	Endurance (Cycles)	ON/OFF ratio	Ref.
Ag/4H-SiC/Pt	3.225×10^{-2}	-	-	$> 10^5$	-	-	[92]
Cu/SiC/W	-	-	5	10^3	100	$> 10^3$	[94]
SiC/Si	-	-	8.5	-	10^9	10^2	[95]
Ag/ZnO@ β -SiC/Pt	10^5	10^3	0	10^4	10^4	25	[96]
ITO/ZnO@ β -SiC/Ag	-	-	-	10^4	10^4	10^3	[97]
Cu/SiC/Pt	-	-	-	-	150	10^3	[98]
Ti/Cu/SiC/Pt	-	-	-	$> 10^3$	> 200	$\sim 10^3$	[93]
Al/TiSbTe/ α -SiC _{0.11} :H/Si ⁺	-	10^2	0	10^4	10^3	-	[91]
Al/Ag NPs: α -SiC _{0.11} :H/Al	-	-	-	10^4	10^4	10^7	[99]

e) Ga₂O₃

The core RS mechanisms of Ga₂O₃-based memristors are centered on oxygen vacancy (*Vo*) migration, predominantly falling into two categories. The first is the CF mechanism, where nanoscale conductive channels are formed by the aggregation of *Vo* or metal cations (e.g., Ag⁺), with the transition between HRS and LRS achieved through CF formation (SET process) and rupture (RESET process), featuring fast switching speeds and high on/off ratios. The second is the ionic-electronic conduction mechanism, in which oxygen ion migration across a broad region modulates the bulk conductivity of the film, enabling gradual switching characteristics that are well-suited for neuromorphic synaptic applications. In terms of fabrication and device architecture, Ga₂O₃ memristors are mainly fabricated via RF sputtering, pulsed laser deposition (PLD), and electron beam evaporation. The mainstream device configuration is the vertical MIM structure, with electrode materials including inert metals (Pt, Au), active metals (Ag, Cu, Al), and transparent conductive oxides (ITO). Device performance can be optimized by incorporating buffer layers such as graphene oxide (GO) and WO₃, as summarized in Table 4. Regarding performance trends, amorphous Ga₂O₃ thin films exhibit low leakage current and superior on/off ratios (up to 10^8), while β -Ga₂O₃ annealed at 600-800 °C demonstrates enhanced stability [100]. Recently, He *et al.* [101] successfully constructed vertical Ag/ β -Ga₂O₃/Pt memristors on high-crystalline-quality β -Ga₂O₃ films via lattice epitaxy engineering and sacrificial layer-assisted transfer, achieving an on/off ratio exceeding 10^8 . Using a scalable polymer-assisted deposition method, Dai *et al.* [102] prepared ferromagnetic amorphous gallium manganese oxide (*a*-GMO) films with a tunable Mn-dopant two-level system, the resulting Pt/*a*-GMO/Pt memristors delivered a on/off ratio of up to 10^3 , benefiting from the synergistic effect of *Vo*-induced LRS and Mn²⁺-enhanced HRS. Suman *et al.* [83] deposited cesium-doped Ga₂O₃ (Cs-Ga₂O₃) nanostructured films via a one-step wet chemical method (Figure 8c), the presence of metal ions near the bottom electrode facilitates rapid CF formation, while Cs doping enables spontaneous diffusion of vacancies in the insulating layer, reducing the *Vo* concentration below a critical threshold to induce CF rupture, thus yielding a retention memory window of 10^5 s. Despite these advancements, Ga₂O₃ memristors generally exhibit cycling endurance of 10^2 - 10^3 cycles and data retention of 10^3 - 10^4 s, along with capabilities for multi-level and analog switching. However, their endurance remains inferior to that of mature oxide-based materials, necessitating further structural optimization for performance enhancement. The performance comparisons in Ga₂O₃-based memristors are listed in Table 4.

Table 4. Performance comparisons in Ga₂O₃-based memristors.

Memristors	Power Consumption (pJ)	Memory window (s)	Forming voltage (V)	Retention memory window (s)	Endurance (Cycles)	ON/OFF ratio	Ref.
Ag/Ga ₂ O ₃ /Au	-	10 ³	2.4	10 ⁴	$> 1.1 \times 10^4$	$> 10^3$	[103]
Al/Ga ₂ O ₃ /Pt/Ti	-	-	4	-	4×10^3	$> 10^5$	[104]
Ag/Ga ₂ O ₃ /Pt	-	-	4.9	10 ⁶	-	-	[100]
Ag/ β -Ga ₂ O ₃ /Pt	-	-	0.5	4×10^4	125	$> 10^8$	[101]
Pt/ α -GMO/Pt	10 ⁵ ~10 ⁶	10 ³	-	10 ³	150	10 ³	[102]
Au/Ti/ β -Ga ₂ O ₃ /Ti/Au	-	10 ²	-	10 ³	200	10	[105]
Pt/ α -GaO _x /ITO	-	-	-	-	-	10 ²	[106]
Al/ α -Ga ₂ O ₃ /ITO	-	10 ²	-8	10 ⁴	100	10 ³	[107]
Ag/Cs-Ga ₂ O ₃ /FTO	-	10 ²	-	10 ⁵	-	10	[83]
Al/GO/ α -Ga ₂ O ₃ /ITO	-	10 ²	-	1×10^2	100	10 ³	[108]
Cu/TiW/Ga ₂ O ₃ /P	-	> 10	-	$> 10^5$	> 140	10 ²	[109]
Ga/GaO _x /SiO _x /p-Si	-	-	-	10 ⁸	10 ⁵	10 ²	[110]
Pt/Ga ₂ O _{3-x} /SiC/Pt	-	10 ²	-	1×10^2	-	10 ³	[36]
Ga ₂ O ₃ /ZnO composite films	-	-	-	10 ⁴	100	4.0×10^2	[111]
Cu/TiW/IGZO/Ga ₂ O ₃ /Pt	-	10 ⁴	5.1	10 ⁴	$> 10^3$	$> 10^3$	[112]

f) Diamond

Diamond-based memristors exploit unique sp²-sp³ carbon phase transitions and defect dynamics. As listed in Table 5, Battistoni *et al.* [113] reported Ti-doped diamond memristors in which RS arises from dehydrogenation/hydrogenation-induced sp²-sp³ transitions, enabling reproducible short- and long-term synaptic plasticity. Liu *et al.* [114] developed an ultra-low threshold voltage responsive synaptonic memristor based on a Cu_xO and bromine-doped diamond (BDD) heterojunction. The HRS and LRS corresponding to oxygen vacancy clusters form the basis of its hysteretic behavior. Synaptic functionality remains observable even when pulse amplitude is reduced to an ultra-low value of 3 mV, with energy consumption as low as 9.5×10^{-11} J.

Table 5. Performance comparisons in diamond-based memristors.

Memristors	Power Consumption (pJ)	Memory window (s)	Forming voltage (V)	Retention memory window (s)	Endurance (Cycles)	ON/OFF ratio	Ref.
Titanium doped diamond	-	-	-	$> 10^5$	> 150	10 ² - 10 ³	[113]
Cu _x O and B-doped diamond (BDD) heterojunctions	95	-	-	-	-	-	[114]

3.2.2 Optoelectronic devices

The broken inversion symmetry of wurtzite GaN gives rise to pronounced spontaneous and piezoelectric polarization along the c -axis, which critically governs its optical and electrical responses. In optoelectronic synaptic devices, polarization-induced built-in electric fields generate potential wells that regulate carrier trapping, separation, and release, thereby enabling history-dependent conductance modulation. Integrating GaN with other strongly polar or wide-bandgap materials has thus emerged as an effective strategy to realize polarization-modulated optoelectronic memristive behavior.

Hua *et al.* ^[115] demonstrated an ultraviolet (UV) optical synaptic device based on a GaN/AlN periodic heterostructure, exhibiting persistent photocurrent (PPC) originating from strong polarization fields at GaN/AlN interfaces. Xie *et al.* ^[116] fabricated a dual-port non-volatile reconfigurable electro-optic dual-mode memristor based on an n-GaN/i-GaN/AlScN heterojunction, where the i-GaN and AlScN layers govern the photoresponsive and storage properties, respectively. The device achieves an ON/OFF ratio of $\sim 10^4$ in both electrical and optoelectronic modes, with reliable resistance state resetting, writing, and long-term retention. Nanostructured GaN platforms further enhance optoelectronic plasticity by amplifying polarization and surface-state effects. As shown in Figure 9a, Ji *et al.* ^[117] prepared amorphous oxide-modified GaN NW arrays ($\text{GaO}_x\text{@GaN}$ NWAs) on flexible graphite paper, exhibiting an ultra-long decay time of 2.3×10^6 s. Similarly, Liu *et al.* ^[118] proposed a photoelectrochemical synaptic device based on p-AlGaIn/n-GaN NWs, internal p-n junctions formed during NW growth on n-Si substrates modulate carrier behavior, and two opposing built-in electric fields enable hole storage in NWs, leading to history-dependent synaptic behaviors. Beyond binary synaptic emulation, advanced bandgap and polarization engineering enables functional diversification. Gao *et al.* ^[119] reported a multifunctional optoelectronic synapse with an n-GaN/i-GaN/AlN multiple-quantum-disk (MQD)/n-GaN architecture. Under 254 nm illumination, efficient carrier confinement by AlN barriers and restricted carrier separation within the n-i-n structure allow the device to emulate both parvocellular and magnocellular pathways in the human retina, enabling contrast-sensitive and motion-sensitive visual processing within a single hardware platform.

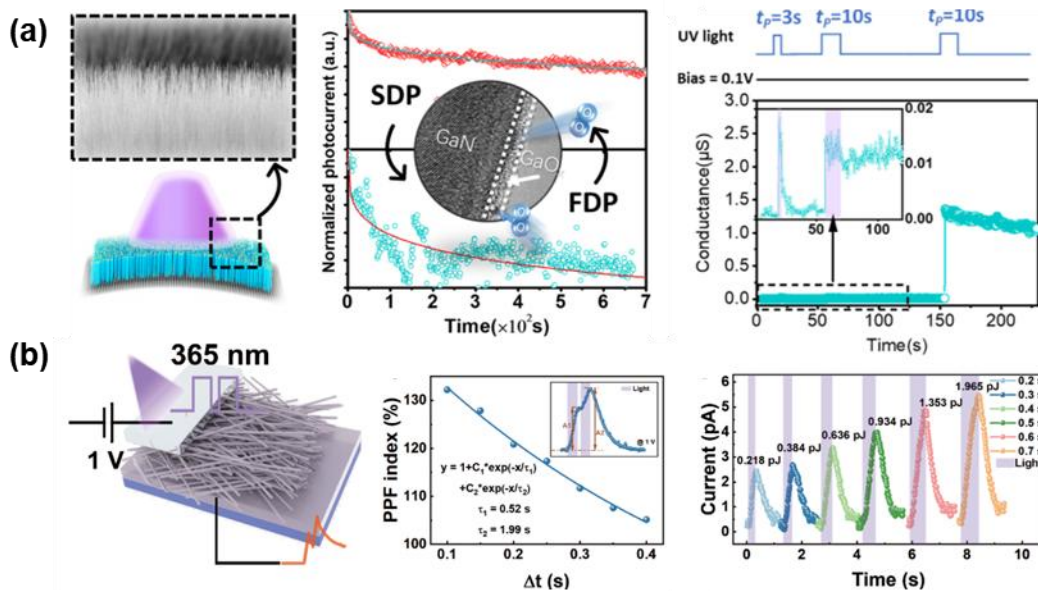


Figure 9. Optoelectronic memristors based on (a) $\text{GaO}_x\text{@GaN}$ NWAs on flexible graphite paper, Reproduced with permission ^[117]. Copyright 2022, Applied Materials. and (b) $\text{SiC/SiO}_2\text{@Ag}$ NW network device. Reproduced with permission ^[120]. Copyright 2025, Science China Materials.

Optoelectronic memristive behavior has also been demonstrated in other WBG systems. Chen *et al.* ^[120] realized a SiC/SiO₂@Ag NW network device capable of UV-responsive synaptic behavior under zero-bias illumination and electrically driven synaptic modulation under external bias (Figure 9b). Benefiting from local field enhancement and localized surface plasmon resonance (LSPR) in Ag nanoparticles, the device operates at ultralow energy consumption (~0.471 fJ per synaptic event). Zhang *et al.* ^[121] fabricated a monolithic 4 × 4 array of vertical Schottky barrier diodes on heavily doped 4H-SiC substrates, where defect-assisted carrier dynamics enable programmable synaptic responses under visible light excitation. Feng *et al.* ^[122] prepared a GaN/Ga₂O₃/GaN back-to-back double heterojunction device, integrating fast photoresponse and PPC effect by regulating the ionization and deionization of V_o in Ga₂O₃. Finally, Mizuno *et al.* ^[123] first demonstrated a photomemristor based on vertically aligned graphene (VG)/diamond heterojunctions. Owing to redox reactions occurring at the graphene layer and/or graphene/diamond (sp²/sp³ carbon) interface, the device achieved light-controlled EPSC, STM-LTM conversion, and PPF with a photovoltaic coupling index exceeding 300%. The device array further functions as an image-sensing memory capable of selective optical information storage. The performance comparisons in WBGs-based memristors and two terminal synaptic devices are listed in Table 6 and Table 7.

Table 6. Performance comparisons in WBGs-based optoelectronics memristors.

Memristors	Wavelength optical power	Power consumption (pJ)	Memory window (s)	Forming voltage (V)	Retention memory window (s)	Endurance (Cycles)	ON/OFF ratio	Synaptic/neuronal functions	Ref.
n-GaN/i-GaN/AlScN	355nm	-	-	-	-	-	10 ⁴	-	[116]
p-AlGaN/n-GaN NW	255 nm 12 μW/cm ²	-	60	-	1.5 × 10 ⁴	-	-	-	[118]
n-GaN/i-GaN/AlN MQDs/n-GaN	254 nm 0.553 mW/cm ²	-	0.85	-	32.57	-	-	-	[119]
GaO _x @GaN	365 nm 8.4 mW/cm ²	-	-	-	2.3 × 10 ⁵	-	~10 ³	-	[117]
InSe/GaN	300-780 nm 40 mW/cm ²	-	-	-	-	-	10 ³	-	[124]
SiC/SiO ₂ @Ag NWN	365 nm 46-246 mW/cm ²	0.471 × 10 ⁻³	0.4	-	0.9	-	-	-	[120]
Ag/SiC/SiO ₂ @Ag(N WN)/ITO	365 nm 326 mW/cm ²	5.936 × 10 ⁻³	-	-	0.9	-	-	-	[120]
Cu/TiW/Ga ₂ O ₃ /Pt	254 nm 130 μW/cm ²	-	> 10 ⁵	-	-	> 140	-	-	[125]
Pt/Ga ₂ O ₃ /Pt	365 nm 0.067-0.471mW/cm ²	-	10 ⁴	5	-	10 ⁴	> 10 ⁴	-	[126]
Ag/Zn-doped Ga ₂ O ₃ (ZGO)/Ag	254 nm 15.70 μW/cm ²	28 × 10 ⁻³	-	-	-	-	-	-	[127]

Memristors	Wavelength optical power	Power consumption (pJ)	Memory window (s)	Forming voltage (V)	Retention memory window (s)	Endurance (Cycles)	ON/OFF ratio	Synaptic/neuronal functions	Ref.
ITO/a-Ga ₂ O ₃ /Pt	254 nm 10 μ W/cm ²	-	-	-	> 3600	10 ⁶	10 ³	-	[128]
Ti -Ga ₂ O ₃ NWs -Au	255 nm 0.16 mW/cm ²	23.9	-	-	-	-	-	-	[129]
Ag/ β -Ga ₂ O ₃ microwire/Ag	254 nm 1000 μ W/cm ²	-	-	-	-	-	10 ³	-	[130]
M1/CNW-Diamond /M2	-	-	-	-	1.374×10^4	-	-	-	[29]
NUNCD-Si/NUNCD -SCD	400-900 nm 1.1-4.3mW/mm ²	-	-	-	-	> 800	-	-	[131]
Ni/Vertical Graphene (VG)/B-doped Diamond	470 nm 78 μ W/cm ²	0.44	5.4	-	1.558×10^3	-	10 ²	-	[123]
Ag/ITZO/Ga ₂ O ₃ /ITO	275-550 nm 40~225 μ W/cm ²	-	-	-	-	-	-	LTD/LTP PPF/PPD STP/LTP	[132]
Ag/Ga ₂ O ₃ /MoS ₂ /ITO	365 nm 0.110~3.290 mW/cm ²	-	-	-	> 500	$> 8 \times 10^3$	-	PPF/PPD STP/LTP STM/LTM	[133]
Ag/Ga ₂ O ₃ /Pt	254 nm 0.8~2.4 mW/cm ²	61.36 544.89	-	3	$> 1 \times 10^4$	80	6.1×10^5	PPF SIDP SNDP STDP SFDP STM/LTM	[134]

Notes: STP represents short-term plasticity, PPF represents paired-pulse facilitation, PPD represents paired-pulse depression, LTP represents long-term potentiation, LTD represents long-term depression, STDP represents spike-timing-dependent plasticity, SRDP represents spike-rate-dependent plasticity, EPSC/EPSP represents excitatory postsynaptic currents/potentials, IPSC/IPSP represents inhibitory postsynaptic currents/potentials, SIDP represents spike intensity-dependent plasticity, SNDP represents spike number-dependent plasticity, SFDP represents spike frequency-dependent plasticity, STM represents short-term memory, LTM represents long-term memory, PPC represents persistent photocurrent, QCSE represents quantum-confined Stark effect

Table 7. Performance comparisons in WBGSSs-based two-terminal synaptic devices

Type	Materials	Device size	Stimulus signal	Spike width (s)	Per spike energy consumption (pJ)	Synaptic/neuronal functions	Refs.
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Planar	AlN/sapphire templates	505 $\mu\text{m} \times$ 560 μm	pulse Xe flash lamp	5×10^{-10}	-	EPSC	[135]
			50-200 Hz			IPSC	
			$V_{DS} = 30 \text{ V}$			PPF	
						STP/ LTP	
Vertical	p-i-n GaN/AlN	-	UV	0.2	-	LTD	[115]
			365 nm			EPSC	
			-40V			PPF	
						STM/LTM	
-	Ti/GaN NWs/Au	-	UV	1	2.72	STDP	[136]
			365nm			EPSC	
			5 V			PPF	
						STM/LTM	
-	p-AlGaIn/n-GaN NWs	-	255nm	0.2	-	STP/LTP	[118]
						PPF	
						STP/LTP	
						PPF	
-	p-GaN/n-AlGaIn/n-GaN NWs	-	265nm	2.5	Self-powered	STP/LTP	[137]
						PPF	
						EPSC	
						PPC	
Vertical	AlGaIn/GaN/Au	-	340 nm	1	-	PPC	[138]
			2V			PPF	
						PPF	
						PPF	
vertical	n-GaN/MQDs/n-GaN	-	254 nm	1	-	SNDP	[119]
			1 V			SRDP	
						QCSE	
						STM-to-LTM	
Vertical	Ti/Al/Ti/4H-SiC/ Ti/Al/Ti	-	405 nm	0.3	5.6×10^4	EPSC	[121]
			2 V			PPF	
						STP/LTP	
						EPSC	
-	Ag / Ga ₂ O ₃ / ZnO / ITO	-	DUV	2	6.64×10^7	PPF	[139]
			254 nm			STP/LTP	
			1 V			EPSC	
						PPF	
-	Ti/Au/ β -Ga ₂ O ₃	3 $\times$ 3 rray	DUV	0.5	5×10^3	STP	[140]
			10 nW/cm ²			PPF	
						PPC	
						STM/LTM	
-	Au/ α -Ga ₂ O ₃ /Si/In	-	254 nm	2	2.36×10^6	EPSC	[141]
			1V			PPF	
						PPC	
						STM/LTM	
-	ITO/ α -Ga ₂ O ₃ /ZnO/ITO	W,L=2 mm	DUV	0.05	0.316	EPSC	[142]
			265 nm			PPF	
			0.6V			STP/LTP	
						EPSC	
-	GTO/Al/HfO _x	W=1000 μm L=30 μm	DUV	0.5	390	PPF	[143]
			254 nm			LTD/LTP	
			0.5 V			SRDP	
						STP	
-	Ti/ β -Ga ₂ O ₃ /Au	W=5 μm L=5 μm	DUV	0.2	2.5×10^{-2}	LTP/LTD	[144]
			258 nm				

Vertical	Ni/VG/Diamond	/	470 nm 8 V	0.05	440	EPSC PPF STM/LTM	[123]
-	Ti/GaN/Ga ₂ O ₃ /GaN/Au	-	254nm 5 V	1	584	EPSC PPF STP/LTP	[122]
-	p-n 4H-SiC	W=25μm L=500μm	375 nm 5 mV	20 ms	0.23	EPSC PPF STDP SRDP SNDP	[145]
-	Pt/Ga ₂ O ₃ /Pt	-	365-660 nm 0.5V	0.5s	-	PPF STM/LTM	[146]

3.2.3 Ferroelectric devices

Ferroelectric materials are intrinsically well suited for neuromorphic computing due to their non-volatile polarization states, low switching energy, and ultrafast response times. By electrically reversing spontaneous polarization, ferroelectric layers can modulate carrier accumulation, depletion, and transport barriers in adjacent semiconductor channels, thereby enabling analog, non-volatile SW storage. In typical metal-ferroelectric-metal (MFM) junctions, polarization charges at ferroelectric/metal interfaces are partially screened by free carriers in the electrodes. Asymmetric screening lengths at the two interfaces result in unequal electrostatic potentials, giving rise to polarization-dependent energy barriers. Assuming different Thomas-Fermi screening lengths ($\delta_1 > \delta_2$) for electrodes M_1 and M_2 , the electrostatic potential φ varies asymmetrically across the junction (Figure 10a), and polarization reversal directly reshapes the transport landscape, switching the device between high- and low-resistance states. Depending on ferroelectric thickness and barrier profile, charge transport may proceed via direct tunneling, Fowler-Nordheim tunneling, or thermionic emission (Figure 10b-d). The performance comparisons in WBGs-based ferroelectric memristors are listed in Table 8.

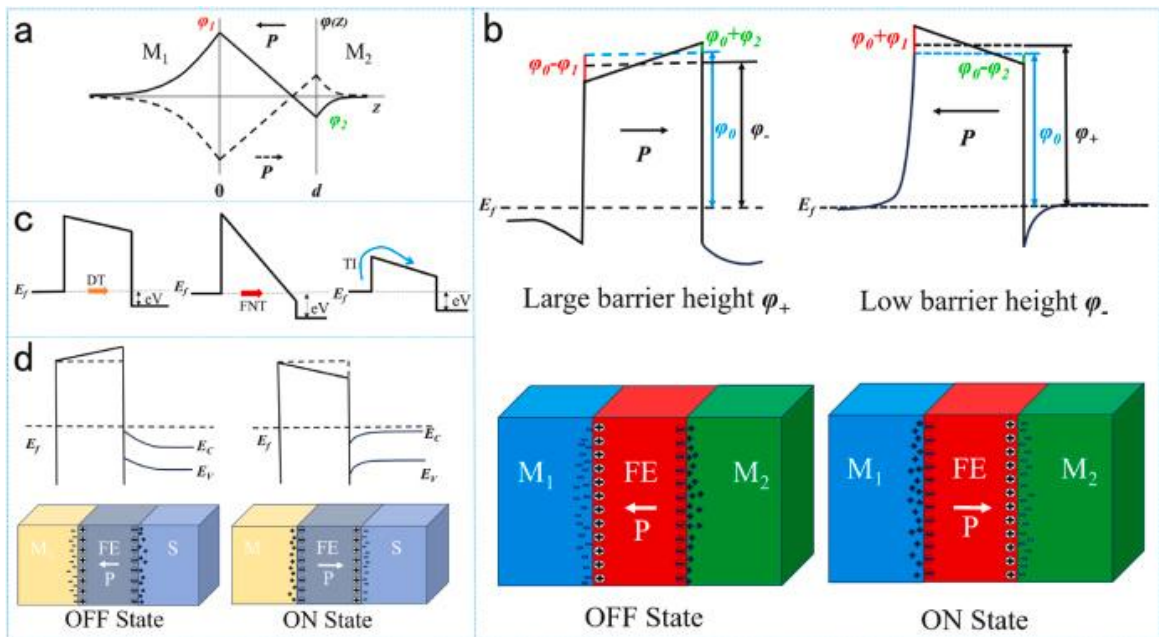


Figure 10. Schematic of ferroelectric polarization-controlled resistive switches. (a) Corresponding asymmetric electrostatic potential distribution (solid line). dashed lines indicate potential when polarization P switches in the ferroelectric device (assuming shielding length L_1 of M is greater than L_2). (b) Large low-ferroelectric tunnel barriers along different polarization directions in the MFM structure, yielding “off” and “on” states respectively. (c) Three possible transport mechanisms demonstrated via ultrathin ferroelectric layers: direct tunneling (DT), Fowler-Nordheim tunneling (FNT), and thermionic emission (TI). (d) Construction of a ferroelectric tunnel junction using the metal-ferroelectric-semiconductor (MFS) structure and corresponding band diagrams. Reproduced with permission ^[147]. Copyright 2025, Materials Today Physics.

In recent years, memristors based on AlN/ScAlN systems and 2D vdW ferroelectric materials have emerged as research hotspots in storage and neuromorphic computing, with numerous high-performance devices successfully developed and their application potential validated. 2D vdW ferroelectric materials attract significant attention due to their electrical tunability, low-power switching, and strong light-matter interactions. Their layered structure facilitates fabricating stable monolayer/few-layer samples, offering an ideal platform to investigate lattice dimensionality effects on long-range ferroelectric order. Integrating these materials into GaN high-electron-mobility transistors (HEMTs) enables low subthreshold swing and advanced neuromorphic computing.

In 2022, Liu *et al.* ^[148] designed an AlScN ferroelectric diode-based memristor, realizing 4-bit pulse-programmable storage and laying the foundation for novel ferroelectric materials in compute-in-memory (CIM) platforms. Concurrently, the Wang group ^[149] developed a fully epitaxial ScAlN/GaN heterojunction device, exhibiting high rectification ratio, long retention, and excellent endurance thanks to its ultra-wide bandgap and high Curie temperature. Its RS relies on ferroelectric polarization-modulated interfacial carrier concentration: positive pulses induce downward polarization (charge accumulation, ON state), while negative pulses trigger upward polarization (electron depletion, OFF state). To address insufficient ON/OFF ratio and high operating voltage in early devices, the Mi group ^[76] tuned ScAlN thickness to 9 nm, leveraging the synergy between its native oxide layer and ferroelectric polarization to achieve a 10^4 - 10^5 ON/OFF ratio and low read/write voltages (<3 V/8 V).

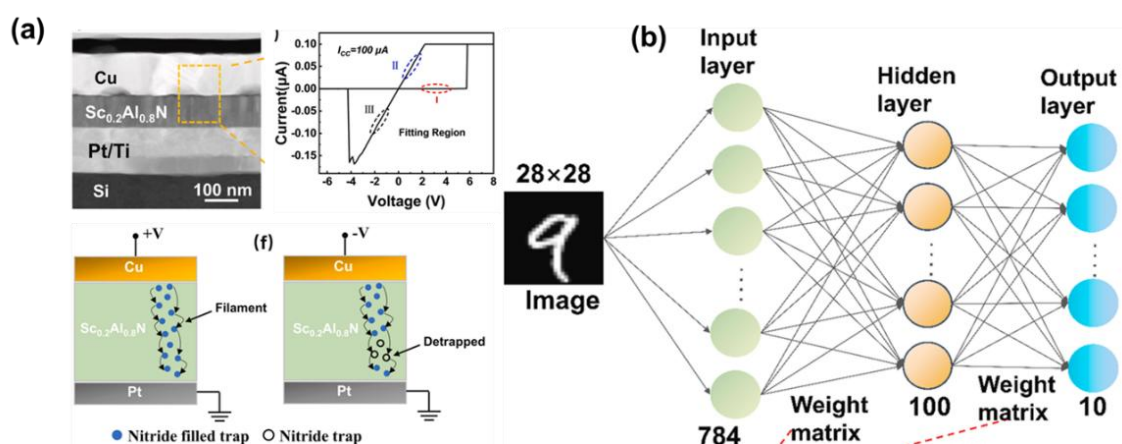


Figure 11. (a) The Cu/Sc_{0.2}Al_{0.8}N/Pt/Ti device and switching process during SET process, (b) Schematic diagram of memristor-based MLP neural networks. Reproduced with permission ^[77]. Copyright 2026, Materials Science in Semiconductor Processing.

Beyond single-layer ferroelectrics, structural and material innovations have further expanded device capabilities. Han *et al.* ^[77] proposed a Cu/Sc_{0.2}Al_{0.8}N/Pt/Ti MIM memristor with defect-assisted conduction. It demonstrated an ON/OFF ratio $> 10^6$, retention > 10000 s, and cycling stability > 200 cycles, enabling multi-level

switching, emulating synaptic behaviors (LTP, PPF), and completing MNIST handwritten digit recognition (Figure 11). Choi *et al.* [72] designed a Pt/Al/TaO_x/AlN/Al₂O₃/Pt device, achieving polarity-dependent dual-mode switching via a 1.2 nm AlN built-in current limiter-abrupt digital switching (forward bias) and gradual analog modulation (negative bias)-with 93.5% classification accuracy on the modified MNIST dataset. To overcome electrical-only control limitations, Xie *et al.* [5] developed an electro-optic dual-mode memristor based on an n-GaN/i-GaN/AlScN heterojunction, synergizing AlScN's ferroelectric storage and GaN's optoelectronic response to switch resistance states via light-modulated polarization reversal threshold.

Finally, the integration of 2D vdW ferroelectrics further enriches WBG neuromorphic platforms. Yang *et al.* [150] integrated α -In₂Se₃ (room-temperature reversible out-of-plane/in-plane polarization) into GaN-based MOS-HEMTs. A self-aligned structure enhanced out-of-plane polarization and suppressed in-plane polarization, achieving a 10 mV/dec subthreshold swing and emulating synaptic behaviors (STP, PPF). Park *et al.* [151] heterogeneously integrated CuInP₂S₆ (CIPS) ferroelectric films as gate dielectrics with AlGaN/GaN HEMTs, realizing SW programming via polarization-induced switching and simulating artificial neural networks (ANNs) through LTP/LTD emulation. These studies advance ferroelectric-based memristors toward high performance, multi-functionality, and low power consumption via material composition regulation, structural innovation, and mechanism exploration, providing crucial support for high-density storage and energy-efficient neuromorphic computing.

Table 8. Performance comparisons in WBGs-based ferroelectric memristors.

Memristors	Power consumption (pJ)	Memory window	Forming voltage	Retention memory window (s)	Endurance (Cycles)	ON/OFF ratio	Ref.
Ni/Al _{0.68} Sc _{0.32} N/Pt	-	-	-	-	2×10^4	1.99	[152]
Ti/AuScAlN/GaN	-	-	-	3×10^5	10^4	60-210	[149]
ScAlN/n-GaN	-	-	-	$> 10^5$	$> 10^4$	10^4 - 10^5	[76]
n-GaN/i-GaN/AlScN	-	-	-	10^3	> 100	10^4	[116]
AlScN/p-i-n GaN	0.088	-	-	$> 10^4$	> 500	9.36×10^5	[5]
GaN/ α -In ₂ Se ₃ HEMT	-	-	-	~ 3.5	$> 5 \times 10^3$	$\sim 10^9$	[153]
CuInP ₂ S ₆ /GaN HEMT	-	-	-	$> 3 \times 10^3$	$> 10^7$	$\sim 1.5 \times 10^8$	[151]

3.3 Synaptic Transistors

Synaptic transistors represent a key class of neuromorphic devices that enable gate-tunable and highly controllable synaptic plasticity. Unlike two-terminal memristors, three-terminal transistor architectures decouple signal transmission and weight modulation, allowing simultaneous read and learning operations with minimal crosstalk. Synaptic transistors based on GaN HEMTs, SiC metal-oxide-semiconductor field-effect transistors (MOSFETs), and related WBG platforms have attracted increasing attention owing to their high carrier mobility, robust thermal stability, and compatibility with advanced semiconductor processing. In these devices, polarization fields, charge

trapping/detrapping at dielectric interfaces, and gate-controlled electrostatic modulation play decisive roles in enabling analog and non-volatile weight tuning. The concept of employing transistors for neuromorphic computing dates back to 1996, when Mead *et al.* ^[154] first demonstrated transistor-based artificial synapses. Since then, synaptic transistors have shown clear advantages in emulating biological synaptic functions, including high-precision and continuous modulation of SW, flexible structural design, and intrinsic compatibility with large-scale integration. Moreover, multi-gate and dual-gate synaptic transistors exhibit structural and functional similarities to biological dendrites, providing a promising route toward advanced synaptic behaviors such as nonlinear integration, spatial summation, and context-dependent learning.

3.3.1 Electronic Transistor

Electronic synaptic transistors based on WBGSs primarily exploit gate-controlled charge trapping, defect-mediated relaxation, and ionic motion to emulate synaptic plasticity. Their three-terminal configuration enables precise and decoupled modulation of channel conductance, providing advantages in controllability and endurance over two-terminal counterparts. As listed in Table 9, Li *et al.* ^[155] proposed an innovative neuromorphic engineering strategy, utilizing unique dendritic dislocations within GaN HEMTs that extend from the bulk material to the channel. Gate voltage (V_G) activates these defect states to trap carriers, while gradual detrapping after bias removal produces a characteristic PSC decay, closely mimicking biological synaptic relaxation. Kim *et al.* ^[156] fabricated fluorine-doped β -Ga₂O₃ thin-film transistors (TFTs) with an ultra-thin channel thickness of ~ 10 nm by controlled reactive ion etching. Electron trapping at the gate dielectric and F- β -Ga₂O₃/SiO₂ interface dominates synaptic behavior, enabling a high on/off ratio (1.61×10^5), low subthreshold swing ($90 \text{ mV} \cdot \text{dec}^{-1}$), and stable analog modulation.

Ionically coupled synaptic transistors further enrich the functionality of electronic devices. Lin *et al.* ^[157] developed a sodium-doped Ga₂O₃ three-terminal electrolyte-gated synaptic transistor (EGST), where sodium ion migration modulates channel conductivity under different gate voltages, emulating the signal transmission mechanism in biological synapses. The EPSC under low positive gate voltage resembles short-term memory (STM), while EPSC under high positive gate voltage simulates long-term memory (LTM). Li *et al.* ^[158] fabricated vertical short-channel Ga₂O₃ TFTs, featuring an ultra-short channel length of ~ 10 nm due to the vertical structure design. This device not only operates stably at low voltage (2 V) with a reasonable on/off of 10^3 but also enables spatiotemporal logic encryption by regulating ion coupling paths. Additionally, heterostructured vdW systems have been explored for programmable synaptic behavior. The Li group ^[159] reported a heterogenized programmable p-n junction based on WSe₂/h-BN/Gr, where part of the WSe₂ channel lies above graphene with a separation of ~ 20 layers of h-BN. Floating-gate charging and discharging can be achieved by applying high drain pulses to the simple two-terminal device, yielding an on/off ratio of $\sim 10^4$ and paving the way for the design of non-volatile memory devices based on vdW heterostructures.

Table 9. Performance comparisons in WBGSs-based electronic synaptic transistors.

Type	Materials	Device size	Stimulus signal (V)	Spike width (ms)	Per spike energy consumpt ion (pJ)	Synaptic/neuronal functions	Refs.
------	-----------	-------------	---------------------	------------------	------------------------------------	-----------------------------	-------

TG	Semiconductor: AlGaIn/GaN Top-gate: Si ₃ N ₄	-	0-10	1	-	EPSC PPF	[155]
SFG	Semiconductor: WSe ₂ /h-BN Semifloating-gate: graphene	-	±40	300	-	PPF	[159]
BG	Semiconductor: F-doped β -Ga ₂ O ₃ Back-gate: thin-film Au	L=450 μ m W=500 μ m	1	0.01	-	PPF LTP/LTD	[156]
BG	Semiconductor: In ₂ O ₃ Back-gate: Na-doped Ga ₂ O ₃	L=100 μ m W=1500 μ m	0.5	20	1.07×10^2	EPSC/IPSC PPF/PPD STM/LTM SNDP LTP/LTD	[157]
coplanar-gates	Semiconductor: α -Ga ₂ O ₃ Gate: sodium alginate (SA)	L=10 nm W=4 mm	2	-	-	Spatiotemporal Logic Encryption	[158]

Notes: BG, TG, SG, SFG and EG represent the bottom gate configurations, top gate configurations, side gate configurations, semifloating gate and extended gate configurations, respectively.

3.3.2 Optoelectronic Transistor

Optoelectronic synaptic transistors introduce optical signal modulation on the basis of traditional electrically controlled synapses, expanding application scenarios. They are mainly classified into two categories: photoelectric synergetic and all-optical types. Their core working mechanism largely relies on defect state-mediated charge trapping and detrapping dynamics-photogenerated carriers are trapped by surface, interfacial, or bulk defects of materials under specific bias voltages, and are slowly released after illumination ceases, resulting in delayed photocurrent decay. This constitutes the key physical foundation for emulating synaptic functions, and this mechanism has been widely applied in the design of optoelectronic artificial synapses and memristive devices. The performance comparisons in WBGSs-based optoelectronic synaptic transistors are listed in Table 10.

GaN-based optoelectronic synaptic devices are a research focus, with a series of high-performance devices successfully developed: The light-stimulated synaptic transistor based on the AlGaIn/GaN HEMT architecture regulates carrier trapping/detrapping dynamics through spontaneous polarization, achieving EPSC responses that vary with optical pulse duration and spike number, thus robustly emulating synaptic functions. Yan *et al.* [160] adopted a 3T III-nitride MOSFET structure on a GaN-on-silicon platform, leveraging the defect charge storage effect at the AlGaIn channel and AlGaIn/SiO₂ interface to exhibit typical synaptic behaviors such as PPF and SRDP.

Su *et al.* ^[161] proposed a dual-channel floating-gate artificial four-color synaptic device with a vertical WSe₂/3D GaN structure, using h-BN as the carrier barrier layer and multilayer graphene (MLG) as the charge storage layer. This device successfully emulates various basic synaptic behaviors including STP, LTP, PPF, SNBP, SRDP, and STP-to-LTP transition, featuring excellent storage performance and broadband response capability. Hong *et al.* ^[162] integrated AlGaIn/GaN HEMT with 2D organic–inorganic halide perovskite (OIHP), realizing gate-tunable positive/negative photoresponses through the light-enhanced field effect, emulating functions such as EPSC, IPSC, and PPF, and achieving full-color perception. The Ni/Au/AlGaIn Schottky-gate GaN-based optoelectronic transistor developed by the Sun group ^[163] eliminates the need for complex or unstable gate dielectrics, enabling self-powered detection, record-high responsivity in photoconductive mode, and optoelectronic synaptic behaviors in the subthreshold region. Its synaptic mechanism originates from the weak carrier depletion effect and PPC effect induced by subthreshold bias. Beyond the HEMT platform, the Sun group achieved a major breakthrough in GaN-based synaptic devices by introducing vdW metal contact as an innovative strategy. Their pioneering work confirmed that the non-destructive nature of vdW interfaces can effectively suppress defect-related carrier recombination, simultaneously achieving high responsivity and PPC effect, thus opening up new possibilities for the design strategy of GaN-based synaptic devices. ^[138]

In addition to GaN-based systems, wide-bandgap material-based optoelectronic synaptic devices have also yielded fruitful results: The 4H-SiC/organic semiconductor (PVK/P3HT) heterojunction synaptic transistor developed by Liu *et al.* ^[164] achieves non-volatility through heterojunction design and optical gating effect, with an energy consumption as low as 0.55 fJ per synaptic event. Its array successfully simulates the dynamic learning and forgetting processes of letter images. The diamond-based neuromorphic retinal perception system (NRPS) proposed by Liu *et al.* ^[26] introduces defects and doping states into diamond thin films through heteroepitaxial growth and silicon doping, realizing broadband spectral response, non-volatile photocurrent, and wavelength sensitivity. It can simulate pupil constriction and image memory, and further accomplish unknown color recognition and dynamic light spot trajectory tracking, improving the UV image recognition accuracy from 67.5% to 81.7%. The nitrogen-doped ultra-nanocrystalline diamond thin film optoelectronic system developed by Yao *et al.* ^[131] adopts an all-wireless architecture. When immersed in physiological media and irradiated with visible (595 nm) or near-infrared (808 nm) pulsed light, charge accumulation at the device-medium interface induces transient ionic displacement currents, thereby achieving electrical stimulation of neurons with high temporal resolution. Du *et al.* applied neuromorphic vision sensors to wide-area diamond quantum sensing, developing an efficient data processing protocol that significantly improves detection speed and precision, laying the foundation for the development of intelligent quantum sensors. Through material integration innovation, structural optimization, and in-depth mechanism exploration, these studies comprehensively promote the development of optoelectronic synaptic devices toward low power consumption, multi-functionality, and high stability, providing diverse and efficient technical support for fields such as neuromorphic computing, artificial vision, and quantum sensing.

Table 10. Performance comparisons in WBGs-based optoelectronic synaptic transistors.

Type	Materials	Device size (μm)	Stimulus signal	Spike width (ms)	Per spike energy consumption (pJ)	Synaptic/neuronal functions	Refs.
TG	GaN/AlN	-	365 nm	100, 200	-	EPSC PPF STM/LTM STDP	[9]
TG	Semiconductor: AlGaIn/GaN	L _G =1	405 nm	200	34.2	EPSC	[165]

	Top-gate: Ni/Au	$L_{GS}=2$ $L_{GD}=3$ $W_G=20$	0.5 V			PPF STM/LTM	
TG	Semiconductor: AlGaIn/GaN Top-gate: SiO ₂	-	375 nm 0.2 mV - 5 mV	60	0.7×10^{-2}	PPF STM/LTM STP/LTP	[160]
TG	Semiconductor: AlGaIn/GaN Top-gate: SiN _x	$W=100$ $L=2$	375 nm 0.001 V	50	0.128	EPSC PPF STM/LTM	[9]
TG	Semiconductor: AlGaIn/GaN Top-gate: SiO ₂	-	350 nm -5 V	100	-	EPSC PPF STP/LTP STP/LTP	[166]
FG	Semiconductor: WSe ₂ /GaN Floating-gate: h-BN/MLG	-	310 nm -8 V	1	-	PPF SNDP SRDP	[161]
TG	Semiconductor: GaN/AlN Top-gate: SiN _x	-	254 nm	50	-	PPF STM/LTM	[119]
TG	Semiconductor: AlGaIn/GaN Top-gate: Al/HfO ₂ /2D OIHP (PEA) ₂ PbI ₄	$W=200$ $L=100$	532 nm 0.1 V	500	-	EPSC IPSC PPF STM/LTM	[162]
TG	Semiconductor: AlGaIn/GaN Top-gate: 2DEG	$W=8.6$ $L=3$	337 nm, 254 nm 10 V	500	-	STP LTP EPSC	[163]
TG	AlGaIn/GaN Top-gate: Ti/Au	-	310 nm, 375 nm, 405 nm 27.1~673 W/m ² V _{gs} : -6~6 V	10, 20, 30	-	EPSC PPF STM-to-LT M STP/LTP	[167]
BG	Semiconductor: Ga ₂ O ₃ /MoS ₂ Back-gate: p-Si	-	532 nm V _{gs} =-25 V	1000	-	PPF STM/LTM	[168]
BG	Semiconductor: Si-doped diamond Back-gate: Si/ SiO ₂	-	From DUV to NIR 222-1064 nm 10 mV	10	222 nm:0.506 532 nm:2.3 × 10 ⁻³	EPSC PPF STDP STM/LTM	[26]

3.3.3 Ferroelectric Transistor

Compared with memristors and optoelectronic synapses, ferroelectric synaptic transistors based on wide-bandgap semiconductors remain relatively underexplored, yet they hold significant promise due to their non-volatile polarization states, fast switching speed, and low energy consumption. To address this gap, Yang *et al.* [153] successfully demonstrated an artificial synapse by integrating a GaN-based MOS-HEMT with α -In₂Se₃, a

ferroelectric semiconductor (Figure 12). The resulting GaN/ α -In₂Se₃ vdW heterojunction is driven by a ferroelectrically coupled two-dimensional electron gas (2DEG), endowing the device with potential for high-frequency operation. Notably, the α -In₂Se₃ semiconductor itself exhibits a steep subthreshold swing and a high on/off ratio of approximately 10¹⁰. A key structural design lies in the synergistic effect between the self-aligned α -In₂Se₃ layer and the gate electrode: this configuration suppresses in-plane polarization while enhancing the out-of-plane polarization of α -In₂Se₃, thereby achieving a steep subthreshold swing of 10 mV/dec and a significant hysteresis of 2 V. Leveraging the STP characteristics of the fabricated ferroelectric HEMT, the researchers further implemented reservoir computing (RC) for image classification tasks. This work highlights the GaN/ α -In₂Se₃ ferroelectric HEMT as a viable pathway toward ultra-fast neuromorphic computing, offering a promising combination of wide-bandgap material advantages, ferroelectric modulation capabilities, and synaptic functionality for next-generation brain-inspired systems.

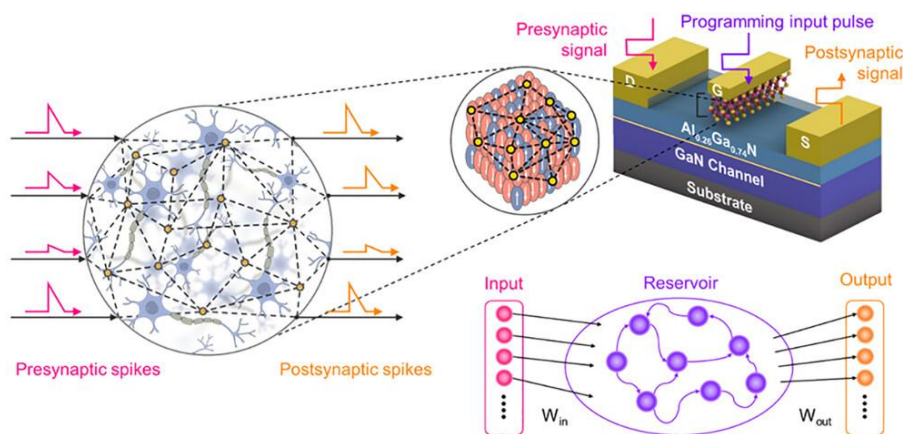


Figure 12. Presynaptic and postsynaptic schematic of the emulated neuron system in RC. Reproduced with permission ^[153]. Copyright 2023, ACS Nano.

3.4 Neuronal Devices

Artificial neuron devices are as indispensable as synaptic elements for constructing complete neuromorphic hardware systems. While numerous mathematical neuron models have been proposed over the past decades, software-based implementations relying on the numerical solution of large sets of differential equations are inherently limited by high computational complexity, substantial energy consumption, and inefficient data movement. In contrast, hardware-implemented artificial neurons directly exploit device-level physical dynamics to emulate neuronal behaviors, offering significant advantages in speed, energy efficiency, and scalability. ^[169] From a device perspective, artificial neurons based on volatile switching materials are especially well suited to emulate biological neuronal dynamics. In such systems, RS thresholds act as spike-triggering conditions, while intrinsic volatility enables spontaneous relaxation, giving rise to oscillatory or pulsed output waveforms analogous to biological spike trains. ^[170] These dynamic behaviors support a wide range of neural network architectures. For instance, coupled neuronal oscillators form the basis of optical neural networks (ONNs), whereas leaky integrate-and-fire (LIF) neurons—characterized by temporal integration, membrane potential decay, and threshold firing—are fundamental building blocks of spiking neural networks (SNNs). ^[171–174] Moreover, the intrinsic stochasticity associated with complex switching physics enables probabilistic spiking behavior, supporting advanced computing paradigms such as Boltzmann machines, Bayesian networks, and pulse-coupled neural networks. ^[175]

A typical biological neuron consists of dendrites, a soma, and an axon (Figure 13). Key neuronal features

include spatiotemporal integration of presynaptic spike trains, short-term memory effects in membrane potential evolution, probabilistic sigmoid-like activation at the axon hillock, and ion-channel-mediated signal transduction across the membrane. Together, these properties allow neurons to aggregate and encode temporally correlated inputs, perform noise-tolerant information processing, and generate rule-based output spike sequences-functional principles that guide the design of artificial neuron devices.

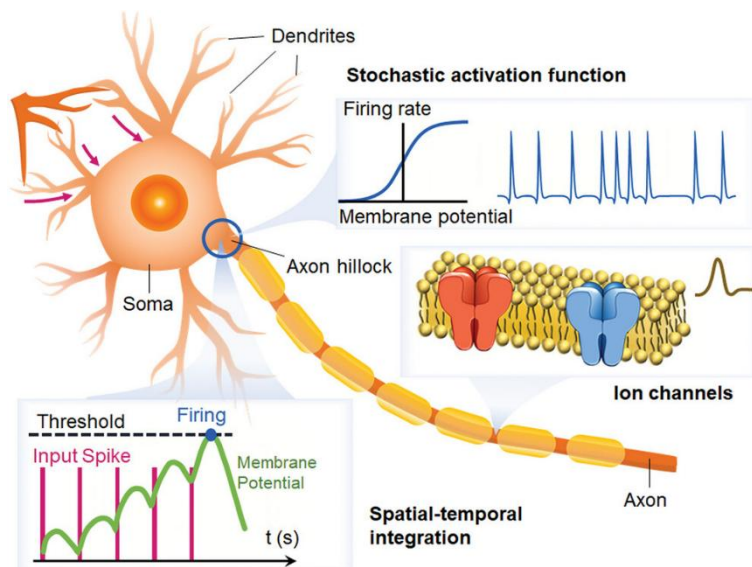


Figure 13. Schematic diagram of a typical neuron. A typical neuron consists of a cell body, axon, and dendrites. The book highlights several key neuronal features: spatio-temporal integration of incoming spike trains, short-term memory effects on membrane potential (lower left diagram), probabilistic sigmoid activation functions in axon hillocks (upper right diagram), and ion channels on the cell membrane (lower right diagram). These properties enable neurons to collect information inputs from multiple synapses, analyze and encode temporal correlations, process information in a fault-tolerant manner, and generate output spike trains (action potentials) according to a set of rules. Reproduced with permission ^[176]. Copyright 2023, Advanced Materials.

To date, reports on neuron devices based on WBGs remain relatively limited. Yan's group ^[177] demonstrated a multilayer Al/AlN/Ag/AlN/Pt thin-film device exhibiting threshold switching behavior with a fast response time of ~50 ns and ultralow leakage current. When incorporated into a LIF circuit, this TS device successfully emulated key neuronal functions, including threshold-triggered spiking, all-or-nothing firing, intensity-dependent spiking frequency, and frequency modulation-surpassing the performance of many earlier artificial neuron implementations. The underlying switching mechanism is attributed to the synergistic migration of nitrogen vacancies and silver filaments, highlighting the potential of nitride-based volatile devices for artificial neuron hardware. In parallel, optoelectronic neuron devices based on WBG materials have emerged as promising biointerfaces. Yao *et al.* ^[131] developed an optically controlled, non-hereditary neural stimulation system using nitrogen-doped ultra-nanocrystalline diamond thin films grown directly on silicon substrates. This all-wireless platform operates under visible (595 nm) or near-infrared (808 nm) pulsed illumination: Charge accumulation at the device-electrolyte interface induces transient ionic displacement currents, enabling high-temporal-resolution electrical stimulation of neurons. The system was validated through photoelectric stimulation of degenerated rat retinas, achieving a photoresponsivity of $7.5 \text{ mA} \cdot \text{W}^{-1}$ under illumination conditions well below ocular safety limits. This work pioneers the use of diamond-based optoelectronic devices capable of generating sufficient photocurrent for direct neuronal activation, underscoring the potential of WBGs in bio-integrated neuromorphic and neuroprosthetic applications.

4. Integration, Benchmarking, and Applications

4.1 Integration with CMOS and Power Electronics

The integration of neuromorphic functionalities with conventional electronics constitutes a central development direction for next-generation intelligent electronic systems. In this context, WBGs have emerged as particularly attractive platforms, owing to their intrinsic material advantages and compatibility with mature semiconductor manufacturing processes. Early efforts to realize neuromorphic behavior in GaN devices focused on emulating human sensory and perception functions. Representative demonstrations include pressure-sensitive GaN HEMTs, in which external mechanical stimuli modulate the 2DEG density and thereby enable real-time regulation of output power, as well as multifunctional devices that integrate pressure sensing and synaptic plasticity by exploiting the strong piezoelectric response of GaN NWs [178]. Although GaN-on-Si heteroepitaxial growth inevitably introduces crystalline defects-such as threading dislocations, stacking faults, and dangling bonds arising from lattice mismatch-these imperfections have been strategically repurposed as three-dimensional charge storage centers for neuromorphic operation. Moreover, the incorporation of piezoelectric, optical, or ferroelectric functionalities has further enriched the neuromorphic response space. Importantly, many of these GaN-based neuromorphic devices are compatible with CMOS fabrication workflows, underscoring their strong potential for large-scale integration and commercialization.

A growing body of work has demonstrated high-performance WBG neuromorphic devices with explicit CMOS and power-electronics relevance. Li *et al.* [155] exploited dendritic dislocations in GaN HEMTs as 3D charge storage sites to emulate neuron-like information processing and synaptic transmission. Notably, these devices were fabricated on 6-inch GaN-on-Si wafers with a yield as high as 95%, highlighting their manufacturability (Figure 14). Sun *et al.* [167] reported a GaN-based optoelectronic integrated chip comprising a series-connected InGaN/GaN multiple-quantum-well (MQW) LED and a GaN optoelectronic synaptic MOSFET, enabling multimodal functionalities such as self-powered detection, wavelength-selective response, and light-emitting synapses. In the ferroelectric domain, Pan *et al.* [179] for the first time grew ferroelectric $\text{Al}_{0.8}\text{Sc}_{0.2}\text{N}$ thin films on CMOS-compatible tungsten substrates, with the resulting memristors exhibiting nanosecond-scale switching speed and excellent retention, the ANN constructed based on these memristors achieved a 93.9% recognition accuracy on the MNIST dataset. Similarly, Wang *et al.* [180] optimized $\text{Al}_{0.77}\text{Sc}_{0.23}\text{N}$ thin films via magnetron sputtering and demonstrated Pd/AlScN/TiN/Si memristors capable of emulating diverse synaptic plasticity behaviors, the corresponding STDP circuit was successfully validated in unmanned aerial vehicle (UAV) flight marker recognition tasks.

Beyond individual devices, complementary integration strategies and emerging materials have further expanded the scope of WBG-based neuromorphic systems. Ren *et al.* [181] developed a CMOS-compatible photonic integrated circuit platform based on AlN/Si waveguides for near-sensor edge computing. By combining the electro-optic response of AlN microring resonators for photonic feature extraction with silicon-based thermo-optic Mach-Zehnder interferometers for neural network computation, this system achieved gesture and gait recognition accuracies exceeding 96%, with sub-10 ns latency and sub-picojoule energy consumption (< 0.34 pJ). Zhu *et al.* [182] integrated h-BN memristors into the BEOL interconnects of CMOS-compatible silicon chips, realizing high-precision current modulation and long cycling endurance, the resulting 1T1M crossbar arrays demonstrated effective power reduction. In parallel, Kapur *et al.* [94] developed memristors based on the BEOL SiC, fabricating Cu/SiC/W stacked devices via CVD. These devices exhibited excellent binary RS with compliance-free and self-rectifying characteristics, offering advantages for high-density 3D crossbar storage architectures and potential for wafer-scale fabrication.

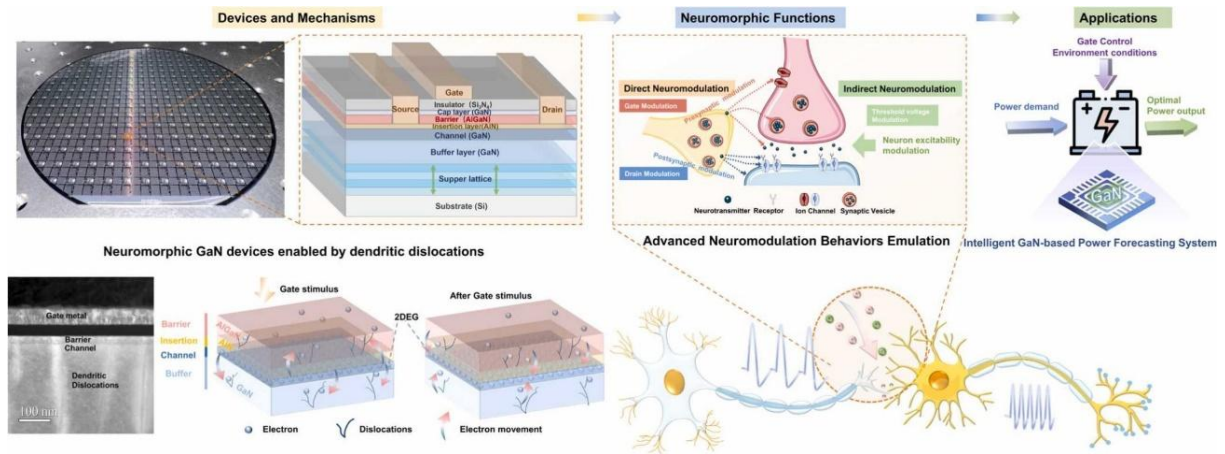


Figure 14. The processing function of neurons emulated by GaN device. Reproduced with permission [155]. Copyright 2024, Nano Energy.

4.2 Application Scenarios

Leveraging their inherent advantages of wide bandgaps, high thermal stability, tunable defect dynamics, and versatile optoelectronic properties, WBGs have spawned diverse high-performance neuromorphic devices. These devices have penetrated key application scenarios of neuromorphic computing, spanning fundamental logic operations, secure signal encryption, brain-inspired visual perception and processing, efficient RC, and extreme-environment high-temperature image learning. The following sections detail representative device architectures, core working mechanisms, and performance benchmarks for each application field, highlighting the unique value of WBGs in advancing intelligent systems.

4.2.1 Logic operation

With the rapid advancement of in-memory computing and neuromorphic systems, there is an urgent demand for multifunctional devices that integrate high-performance storage, logic operations, and synaptic plasticity-capabilities critical to overcoming the von Neumann bottleneck. WBGs have emerged as promising candidates for such applications, leveraging their inherent advantages of ultra-low leakage, high breakdown fields, and tunable defect/interface dynamics.

He *et al.* [101] constructed a vertical Ag/ β -Ga₂O₃/Pt memristor on high-crystalline-quality β -Ga₂O₃ thin films via lattice epitaxy engineering and sacrificial layer-assisted transfer (Figure 15a). The β -Ga₂O₃-based device exhibited exceptional performance, including an ON/OFF ratio exceeding 10⁸, low programming/reset voltages (0.13 V/−0.11 V), ultra-low programming current (10^{−10} A), long data retention (> 4 × 10⁴ s), and excellent subthreshold characteristics (~0.47 mV·dec^{−1}). Tunable I_{cc} enabled the coexistence of volatile and non-volatile switching modes. Electrical characterizations combined with first-principles calculations revealed that the diverse RS behaviors are dominated by Ag ion migration. A reconfigurable non-volatile XOR logic gate circuit based on this β -Ga₂O₃ memristor was designed and simulated, enabling image encryption/decryption and edge detection functions. This work not only validated the feasibility of lattice-engineered high-quality β -Ga₂O₃ thin films for advanced memristor fabrication but also expanded their applications into digital logic and reconfigurable image processing. Cui *et al.* [134] proposed an amorphous wide-bandgap Ga₂O₃ optoelectronic synaptic memristor, achieving 3-bit data storage by modulating I_{cc} and variable ultraviolet (UV-254 nm) light intensity (Figure 15b). By

utilizing voltage polarity and UV light as input signals, the device realized “AND” and “OR” logic gate functions in memristor-aided logic (MAGIC). It also exhibited highly stable synaptic characteristics, including PPF, spike intensity-dependent plasticity (SIDP), spike number-dependent plasticity (SNDP), STDP, spike frequency-dependent plasticity (SFDP), and learning-experience behaviors. When integrated into an ANN, the Ag/Ga₂O₃/Pt memristor simulated light-pulse potentiation and electric-pulse depression with a high pattern accuracy of 90.7%, making this multifunctional single memristive unit a promising candidate for optoelectronic memory, neuromorphic computing, and artificial visual perception applications. Xie *et al.* [116] adopted an n-GaN/i-GaN/AlScN heterojunction Schottky diode structure, achieving an ON/OFF ratio of $\sim 10^4$ in both electrical and optoelectronic modes, with reliable resistance state resetting, writing, and long-term retention (Figure 15c). In the electrical mode, resistance state changes originate from ferroelectric polarization reversal in the AlScN layer of the heterostructure, which modulates the device’s depletion region width and electron transport barrier height. In the optoelectronic mode, the storage window can be tuned by light intensity via the photoconductive effect of the i-GaN layer and light-induced reduction of the electron transport barrier, thereby adjusting the partial bias applied to the AlScN layer. Based on this device, the implication (IMP) truth table and logic “FALSE” state were successfully reproduced, demonstrating its great potential in in-sensor computing and memory-computing-integrated systems.

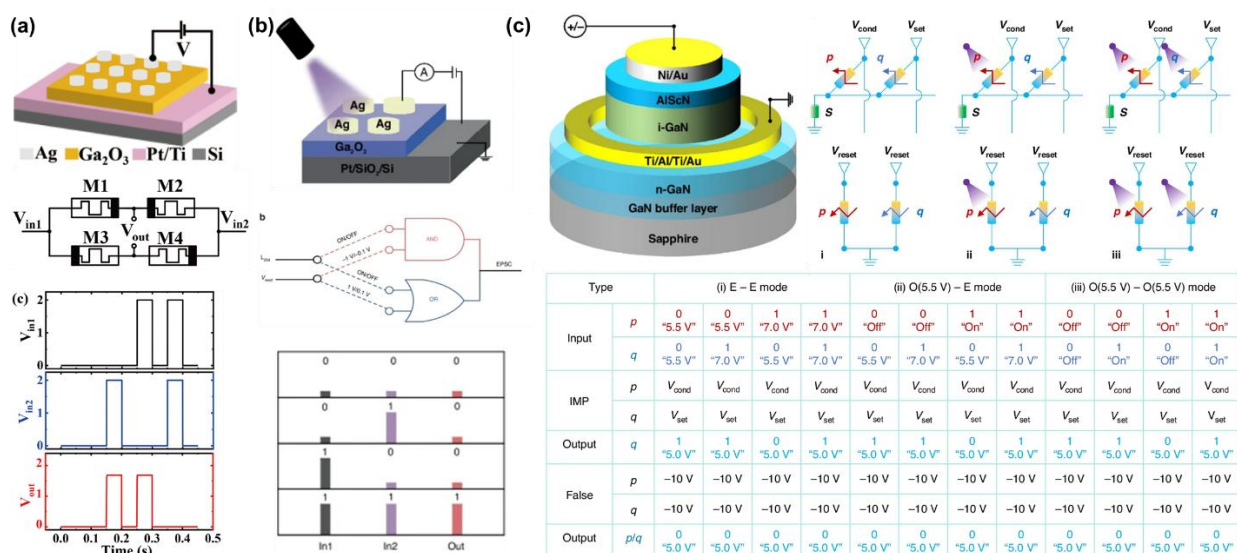


Figure 15. (a) β -Ga₂O₃-based memristor circuit for image encryption, Reproduced with permission [101]. Copyright 2026, Science China Materials. (b) Architecture and Performance of a Ga₂O₃-Based Photoelectric Synaptic Memristor for Logic Gates, Reproduced with permission [134]. Copyright 2025, Light: Science & Applications. (c) The electro-optic duplex memristor on a GaN/AlScN hetero-structure based Schottky diode. Reproduced with permission [116]. Copyright 2024, Light: Science & Applications.

4.2.2 Signal encryption

Currently, signal encryption research based on WBGs primarily focuses on memristors and optoelectronic devices, leveraging the unique electrical, optical, and environmental stability of WBG materials to address emerging security challenges in hardware systems. Li *et al.* [158] developed vertical short-channel Ga₂O₃ TFTs to mitigate security threats in critical hardware components (Figure 16a). Featuring an ultra-short channel length of ~ 10 nm

enabled by the vertical structural design, the device operates stably at a low voltage of 2 V with a reasonable on/off ratio of 10^3 . More importantly, it achieves spatiotemporal logic encryption by regulating ion coupling paths, while its AND-gate logic decryption function can only be activated by reducing the gate-channel spacing-enhancing hardware security through structural design. Shen *et al.* [183] demonstrated the capability of 3C-SiC NWs synaptic devices to distinguish Morse code characters. Their findings revealed that these devices respond to optical signals representing international Morse code, with each character triggering a distinct EPSC amplitude response. Individual characters can be identified by analyzing the sum and endpoints of EPSC amplitude peaks; comparative analysis of SUM and END patterns confirmed accurate recognition of each Morse code, facilitating encrypted data transmission and identification. Liu *et al.* [145] exploited the significant optical gating effect induced by efficient separation of photogenerated carriers at the p-n 4H-SiC homojunction interface, enabling robust light-stimulated synaptic behaviors in the device without encapsulation in ambient air at 350 °C. Through array integration, the devices successfully realized information encryption and image learning/memory functions at 350 °C, highlighting their potential for high-temperature secure computing scenarios. As a fundamental degree of freedom of light, optical polarization provides a powerful dimension to enhance information capacity in optoelectronic systems. By encoding data into polarization states (e.g., linear, circular, or elliptical polarization), a single light beam can transmit multiplexed information channels, offering a superior alternative to traditional intensity-based signals. This unique property has driven advanced applications in optoelectronics, including reconfigurable photonic logic gates based on polarization-sensitive devices, as well as advancements in secure communications where polarization states serve as cryptographic keys for optical information encryption and decryption. Zhao *et al.* [130] obtained high-quality β -Ga₂O₃ (MWs) from bulk single crystals via simple mechanical exfoliation (Figure 16b). Utilizing the inherent polarization-sensitive anisotropy of β -Ga₂O₃, a lens-free polarized photodetector was realized, exhibiting solar-blind operation insensitive to visible light backgrounds, an ultra-high polarization ratio of 137 at 254 nm, and excellent reproducibility. Leveraging this exceptional polarization sensitivity, the study achieved on-chip reconfigurable OR/AND logic operations via polarization modulation and realized optical encryption based on polarization-encoded signals.

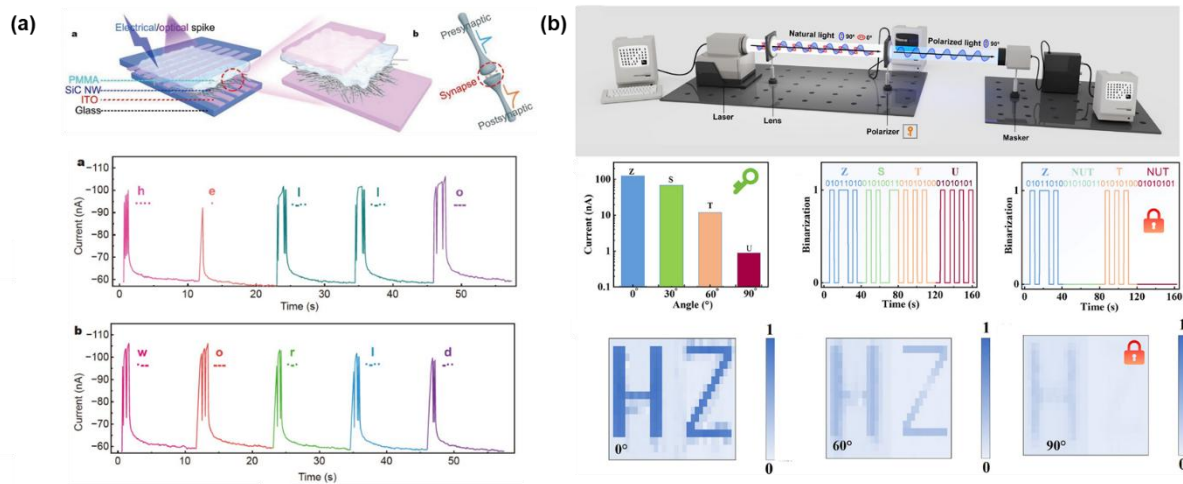


Figure 16. (a) 365 nm light-triggered EPSC amplitudes of SiC NW artificial synaptic device under International Morse code stimulation, Reproduced with permission [184]. Copyright 2023, Science China Materials. (b) Polarized light encryption information transmission system. Reproduced with permission [130]. Copyright 2025, Advanced Optical Materials.

4.2.3 Visual Perception and Processing

Humans acquire over 80% of information about the external environment through the visual system. Efficiently emulating the perceptual capabilities and information processing mechanisms of the human visual system has long been a core goal of bionic research. Currently, WBGs-based neuromorphic devices exhibit remarkable potential in brain-inspired visual perception and processing. In terms of electronic devices, the sodium-doped Ga_2O_3 three-terminal EGST demonstrates high-accuracy pattern recognition capabilities when integrated into ANNs. For optoelectronic devices, all types of WBGs mentioned herein have been applied in image recognition. The current trend is to fabricate device arrays to improve recognition accuracy, thereby enabling the recognition of complex and dynamic images. Sun Group^[137] first reported a p-GaN/n-AlGaN/n-GaN nanowires diode grown on a Si substrate (Figure 17a). This embedded electron reservoir architecture enables sophisticated regulation of charge trapping and detrapping dynamics, breaking the operational limitations of conventional p–n junction paradigms. It endows the system with bias-modulable optoelectronic characteristics, including tunable photoresponsivity reaching 10.45 mA/W, photonic synaptic behaviour with a PPF as high as 122%, and optical memory performance featuring eight discrete linear conductance states. Furthermore, the three-in-one diode array is capable of constructing compact, energy-efficient image sensing hardware. Zhang *et al.*^[168] successfully fabricated a $\text{Ga}_2\text{O}_3/\text{MoS}_2$ heterojunction optoelectronic device by stacking n-type Ga_2O_3 and n-type MoS_2 layers (Figure 17b). The device achieves a rectification ratio of $\approx 10^5$, an on/off ratio of $\approx 10^8$, a high detectivity of 1.34×10^9 Jones, and a high responsivity of 28.92 mA/W. More importantly, this heterojunction device demonstrates potential for integrating sensing and storage functions, making it suitable for use as an artificial neuromorphic synapse. By synergistically regulating the optical switching state and gate voltage, the device exhibits information processing capabilities as the optoelectronic logic gate “AND”.

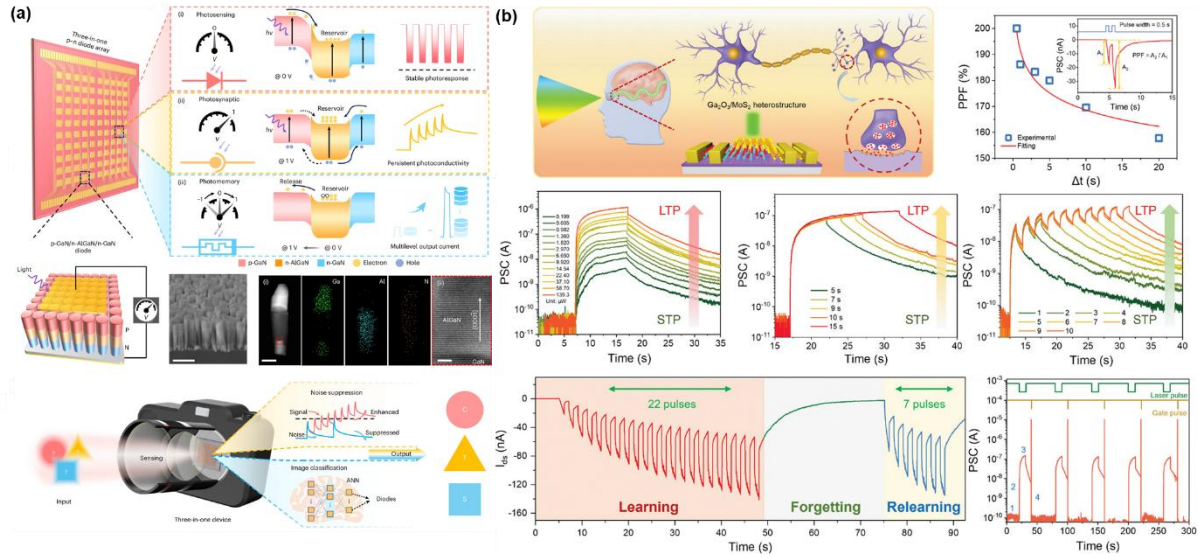


Figure 17. (a) Three-in-one image sensor based on a multifunctional p-GaN/n-AlGaN/n-GaN nanowires diode architecture. Reproduced with permission^[137]. Copyright 2026, Nature Electronics (b) Neuron/synaptic functions of the $\text{Ga}_2\text{O}_3/\text{MoS}_2$ heterostructure device as a neuromorphic vision sensor. Reproduced with permission^[168]. Copyright 2024, Advanced Functional Materials.

4.2.4 Reservoir Computing

RC, a machine learning framework, has garnered significant attention for its ability to circumvent error accumulation in traditional recurrent networks while reducing hardware costs. Within neuromorphic architectures, RC is renowned for its suitability in processing time-series and high-dimensional data, featuring a three-layer structure: an input layer, an untrained fixed reservoir layer that nonlinearly maps inputs to a high-dimensional space, and a trainable linear readout layer for high-dimensional state classification. RC systems based on spintronic oscillators, photonic modules, nanonetworks, and memristors have been successfully demonstrated. Recently, an in-sensor RC system based on GaN HEMTs was proposed to maximize the performance of neuromorphic computing for in-situ processing of large datasets in hardware. Jiang *et al.* ^[185] experimentally validated an in-sensor RC system for intelligent gas sensing, where the core GaN HEMT exhibits nonlinear dynamic channel conductivity that varies with gas concentration. This device follows gate-modulated nonlinear dynamics with a fading memory effect, physically encoding each 4-bit sequence into a unique and distinguishable ID. A single-layer fully connected neural network (FCNN) leverages these reservoir states (IDs) to learn gas pattern classification. Guo *et al.* ^[186] demonstrated a BEOL-compatible SiC-based memristor that exhibits short-term memory properties and can encode temporal signals. A physical RC system centered on this memristor was constructed and experimentally validated for pattern recognition tasks (Figure 18). After training, the RC system achieved 100% accuracy in classifying 0–9 digit patterns and demonstrated excellent robustness against noisy pixels.

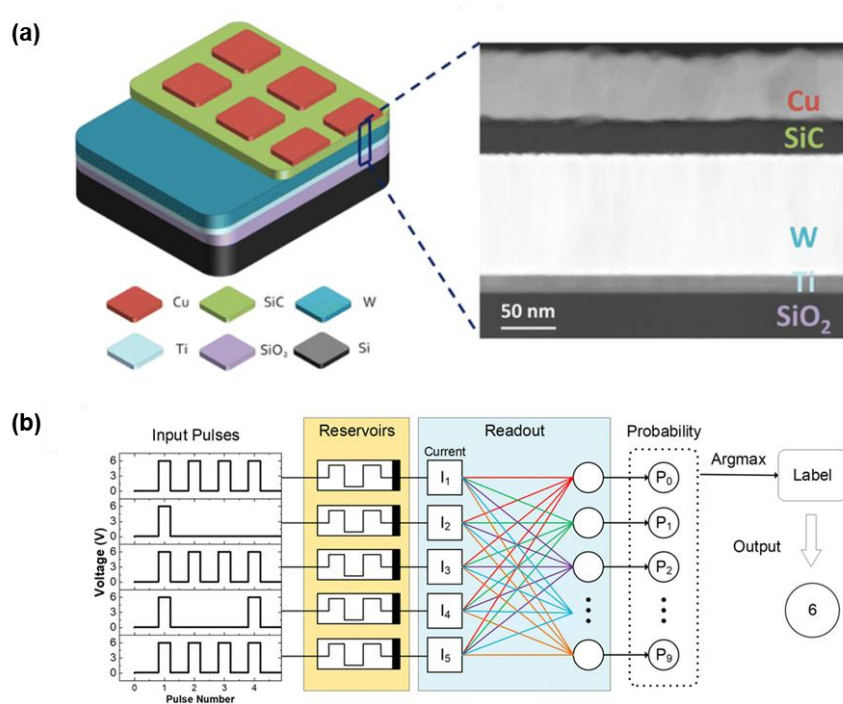


Figure 18. (a) Schematic of the Cu/SiC/W memristor structure, (b) Schematic representation of the physical RC system including the inputs (pulse streams), the SiC memristor reservoir and the readout network. Reproduced with permission ^[186]. Copyright 2023, Materials Advances.

4.2.5 High-Temperature Image Learning and Memory

Most neuromorphic devices currently operate solely at room temperature, limiting their deployment in extreme-environment computing scenarios such as aerospace, automotive, geothermal, and oil & gas industries. For instance, high-temperature image preprocessors in UAVs could capture close-range imagery near volcanoes or wildfires if capable of withstanding extreme temperatures, while also reducing image computation loads. Developing neuromorphic devices for high-temperature front-end data preprocessing thus helps expand application boundaries.

WBGSs inherently tolerate higher temperatures: Ga_2O_3 has shown potential in power devices with a breakdown voltage of 8.32 kV, and recently, in-memory sensing systems based on Ga_2O_3 synaptic memristor arrays were demonstrated for intelligent fingerprint recognition. However, amorphous Ga_2O_3 crystallizes at elevated temperatures, restricting high-temperature applications. Benefiting from both high-temperature resistance and polarization properties, $\epsilon\text{-Ga}_2\text{O}_3$ emerges as a promising candidate. As shown in Figure 19a, Lin *et al.* [187] fabricated a high-temperature image preprocessor using $\epsilon\text{-Ga}_2\text{O}_3$ optical synapses: High-quality $\epsilon\text{-Ga}_2\text{O}_3$ thin films were epitaxially grown via mist chemical vapor deposition, and a large photodetector array (152 devices) was constructed, featuring pA-level dark current, 5-6 orders of magnitude photo-switching ratio, and high uniformity. Notably, the devices exhibited significant photo-induced compression effects and synaptic behaviors, which improved at high temperatures-originating from the competitive mechanism between carrier trapping/detrapping and mobility fluctuation. Based on this, high-temperature image compression was achieved: 4×4 pixel images were compressed to 4×1 vectors at 650 K, reducing data volume for subsequent recognition, with 100% accuracy after 32 training epochs at 650 K.

Other WBG materials have also been explored for high-temperature optoelectronic synapses: Xiaodong Pi's group [188] developed 4H-SiC-based devices, where a 3×3 array-based neural network showed stable performance across temperatures, realizing multiple optoelectronic synaptic functions up to 600 K (Figure 19b). A 4H-SiC p-n homojunction synaptic device achieved superior light-stimulated synaptic behaviors in ambient air at 350 °C (without encapsulation) due to efficient separation of photogenerated carriers at the homojunction interface, enabling information encryption and image learning/memory simulations at 350 °C [145]. Its array achieved 95% accuracy in MNIST handwritten digit recognition via ANN simulation. Shen *et al.* [189] reported high-temperature brain-inspired synaptic devices based on SiC@NiO core-shell NWs network optoelectronic memristors (NNOMs), which exhibited short/long-term synaptic plasticity and modulatory plasticity under electrical/optical stimulation, with stable short/long-term memory and “learning-forgetting-relearning” functions at room temperature and 200 °C. A 5×3 optoelectronic synaptic array maintained stable visual memory at 200 °C, suitable for artificial visual system development. These WBG-based devices demonstrate great potential as high-temperature optoelectronic synapses, expanding the operating temperature range of optoelectronic neuromorphic computing systems.

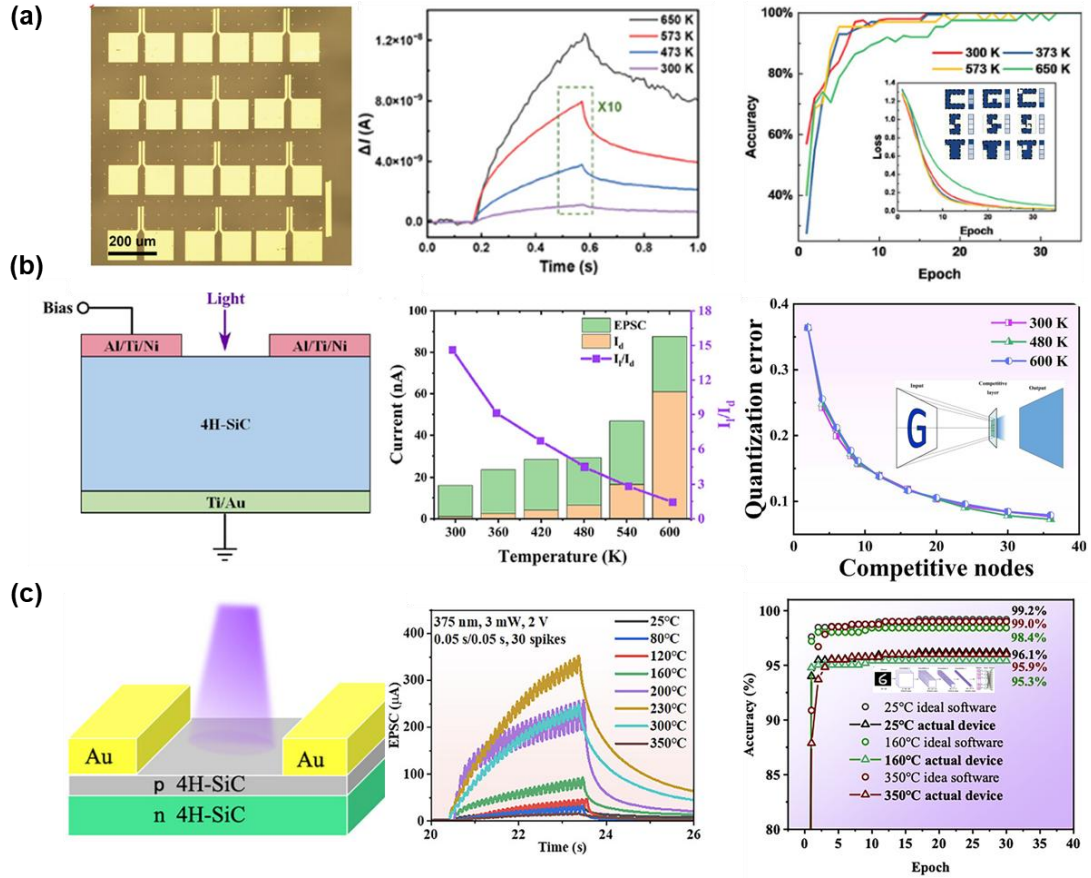


Figure 19. (a) Photodetectors based on ϵ -Ga₂O₃, Reproduced with permission ^[187]. Copyright 2024, Advanced Physics Research. (b) 4H-SiC synaptic device, Reproduced with permission ^[188]. Copyright 2025, Science China Information Sciences. (c) High-temperature-resistant synaptic devices based on the p-n 4H-SiC homojunctions. Reproduced with permission ^[145]. Copyright 2025, Small.

5. Challenges and Future Perspectives

Despite the rapid progress of WBGs-based neuromorphic devices, several critical challenges must be addressed before these materials can be translated into large-scale, reliable, and application-ready neuromorphic systems (Figure 20).

Material Uniformity and Defect Control. Defects play a dual role in WBG neuromorphic devices. While vacancies, dislocations, and interface states are often deliberately exploited to enable analog RS and synaptic plasticity, their stochastic nature also introduces device-to-device variability and cycle-to-cycle instability. This challenge is particularly pronounced in materials such as GaN and Ga₂O₃, where dislocations, oxygen vacancies, and interface traps are strongly dependent on growth conditions and substrate choice. Future research should move from defect utilization toward defect-by-design strategies, in which defect types, spatial distributions, and densities are precisely engineered through advanced epitaxial growth, doping control, and post-processing techniques. Establishing clear correlations between defect physics and neuromorphic functionality will be essential for improving uniformity and reproducibility.

Device Reliability and Energy Efficiency. Although WBG materials inherently offer high breakdown fields and thermal robustness, long-term reliability under repeated programming cycles remains a concern, especially for

defect- or ion-migration-based devices. Issues such as conductance drift, retention degradation, and switching endurance must be systematically evaluated under realistic operating conditions, including elevated temperatures and high electric fields. In addition, some WBG neuromorphic devices still operate at relatively high voltages compared with biological synapses or emerging low-power oxide systems. Reducing operating voltage and energy consumption through interface engineering, polarization-assisted modulation, and ferroelectric integration represents an important direction for future device optimization.

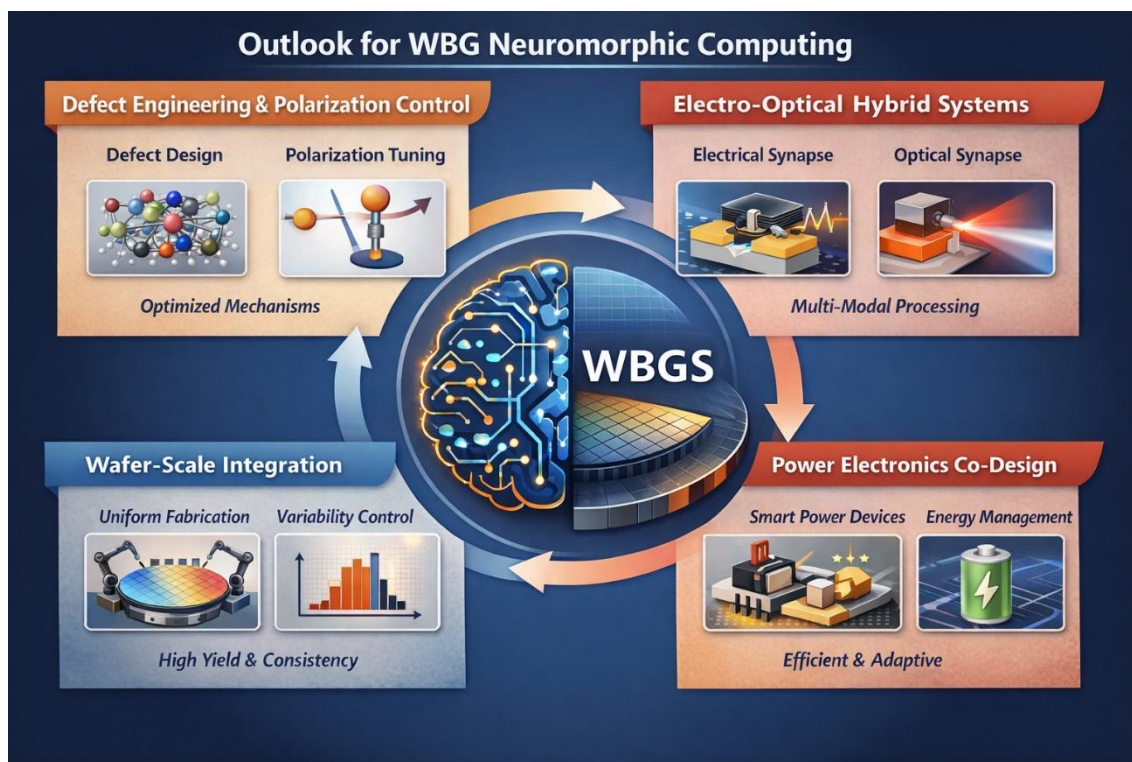


Figure 20. Future Perspectives for WBGs-based Neuromorphic Computing. The author owns the copyright.

Scalability and Integration Compatibility. Scalable integration is a key bottleneck for WBG neuromorphic hardware. While GaN and SiC benefit from relatively mature wafer-scale growth and CMOS compatibility, other materials such as Ga_2O_3 and diamond still face challenges related to thermal management, substrate availability, and process standardization. Ga_2O_3 -based memristors have exhibited favorable compatibility with CMOS and 3D integration in preliminary attempts, yet such integration technology remains in the early developmental stage. Thus, the integration of neuromorphic functionality with existing electronic platforms-particularly BEOL compatibility, three-dimensional crossbar architectures, and hybrid CMOS-WBG systems-requires further development. Advancing low-temperature fabrication processes and heterogeneous integration schemes will be crucial for enabling high-density neuromorphic circuits.

Algorithm-Hardware Co-Design. A lack of standardized benchmarking metrics makes it difficult to fairly compare WBG neuromorphic devices with conventional material platforms. Parameters such as linearity, symmetry, dynamic range, energy per synaptic event, temperature stability, and noise tolerance should be evaluated in a unified manner. Furthermore, most demonstrations to date rely on conventional neural network models that are not optimized for the intrinsic properties of WBG devices. Developing algorithm-hardware co-design strategies-such as SNNs and RC architectures tailored to nonlinear switching, volatility, or optoelectronic coupling-will be essential to fully exploit the unique advantages of WBG materials.

Multifunctional and Extreme-Environment Intelligence. Looking forward, WBGs offer distinctive opportunities that extend beyond conventional neuromorphic computing. Their robustness under high temperature, high radiation, and high electric field environments positions them as ideal candidates for intelligent systems in aerospace, nuclear energy, and geothermal applications. In addition, the strong optoelectronic and polarization properties of WBG materials enable natural integration of sensing, memory, and computing within a single device, opening pathways toward in-sensor and edge intelligence. The convergence of neuromorphic computing with power electronics, photonics, and harsh-environment electronics represents a particularly promising direction for future research.

In summary, wide-bandgap semiconductors are transitioning from passive electronic materials to active enablers of neuromorphic computing. Continued advances in defect engineering, interface design, scalable integration, and algorithm-aware device development are expected to unlock their full potential. As these challenges are progressively addressed, WBG semiconductor-based neuromorphic computing is poised to complement and extend existing technologies, enabling robust, efficient, and multifunctional intelligent systems operating beyond the limits of conventional material platforms.

6. Conclusions

WBGs, including GaN, AlN, BN, SiC, Ga₂O₃, and diamond, have emerged as a powerful materials platform for neuromorphic computing beyond the limitations of conventional silicon-based systems. Their intrinsic properties—large bandgaps, high breakdown fields, excellent thermal stability, rich defect physics, and strong polarization effects—enable robust, energy-efficient, and multifunctional neuromorphic devices capable of operating under extreme electrical, thermal, and radiation environments. This review has systematically summarized recent advances in WBG-based neuromorphic computing from materials fundamentals to device architectures, physical mechanisms, and application demonstrations.

From a materials and device perspective, III-V nitrides enable polarization-modulated and optoelectronic synaptic functionalities, SiC supports high-temperature and radiation-hard neuromorphic systems, Ga₂O₃ offers ultra-high breakdown fields with defect-tunable memristive behavior, and diamond provides superior thermal management and low-noise characteristics for photonic neuromorphic applications. Building on these material advantages, diverse neuromorphic device architectures—including memristors, synaptic transistors, and artificial neurons—have been realized through defect- and interface-mediated transport, polarization-controlled carrier modulation, and photo-electro-thermal coupling, achieving key synaptic and neuronal functions with performance metrics that rival or exceed biological and conventional counterparts. At the system level, WBG neuromorphic devices demonstrate favorable CMOS compatibility, enabling BEOL integration and 3D crossbar architectures. Their successful deployment in logic operations, visual perception, RC, and high-temperature image learning highlights particular strengths in edge intelligence and extreme-environment applications where traditional technologies face fundamental limitations. These advances underscore the unique potential of WBG materials to merge sensing, memory, and computing within robust and compact hardware platforms.

Looking forward, challenges related to defect-induced variability, long-term reliability, large-scale integration, and algorithm-hardware co-design remain to be addressed. Continued progress in defect-by-design strategies, multifunctional device innovation, and neuromorphic architectures tailored to WBG material characteristics is expected to unlock the full potential of this material class. With ongoing advances in material engineering and system integration, WBG semiconductor-based neuromorphic computing is poised to become a key enabler of next-generation intelligent systems operating beyond conventional material and environmental limits.

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Author contribution. H.T. designed this study. H.T., P. M., and Y.Z., collectively conducted the literature review and wrote the manuscript. H.T., Q.Z., W. Z., R. G., and G. Z. provided suggestions and revised the manuscript.

Competing interests. The authors declare that they have no competing interests.

Data Availability. Data available on request from the authors.

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